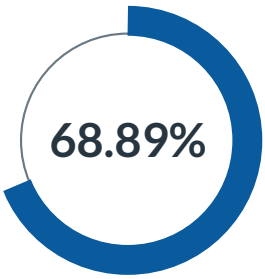
Results

Amanda Kondrat'yev — 1st Attempt



62
Out of 90 points

01:05:58
Time for this attempt

This assessment has unlimited attempts. Take Now

Attempt History

Results	Points	Score	(Highest score is kept)
Attempt 1	62 of 90	68.89%	(Highest score)

Your Answers:

1 1 / 1 point Multiple Choice

Which option would you MOST likely use to configure PXE to set up remote imaging on a computer during a deployment?

- ○ Secure boot configuration
- ✓ ◉ BIOS boot option
- ○ Windows boot menu
- ○ Power management configuration

Feedback

General Feedback

OBJ 3.5: To configure a computer to utilize the PXE boot option, you need to configure it under the BIOS boot option. The Preboot Execution Environment (PXE) is an industry-standard client/server interface that allows networked computers that are not yet loaded with an operating system to be configured and booted remotely by an administrator. Secure boot is a security system offered by UEFI that is designed to prevent a computer from being hijacked by malware. Power management allows Windows to reduce or turn off the power supplied to hardware components to save energy.
For support or reporting issues, include Question ID: 63fe04fa3b7322449ddbaf73 in your ticket. Thank you.

2 1 / 1 point Multiple Choice

Your company's printer is displaying an error of "Replace cartridge 1. Filament is empty." Based on this error, what kind of printer is your company using?

- ○ Thermal
- ○ Inkjet
- ✓ ◉ 3-D
- ○ Laser
- ○ Impact

Feedback

General Feedback

OBJ 3.8: 3-D printers use a filament to create the objects being printed. A 3D printing filament is a thermoplastic feedstock for fused deposition modeling by 3D printers. There are many types of filament available with different properties, requiring different temperatures to print the final product or object. On the exam, you should be thinking about a 3D printer anytime you see the word filament. An inkjet cartridge is used with an inkjet printer. Toner is used with a laser printer. Ribbon is used with an impact printer. Ribbon or special thermal paper is used with a thermal printer.
For support or reporting issues, include Question ID: 63fe05013b7322449ddbafd4 in your ticket. Thank you.

3 1 / 1 point Multiple Choice

Which of the following is a connection-oriented protocol that utilizes TCP?

- TFTP
- DHCP
- SNMP
- HTTPS

Feedback

General Feedback

OBJ 2.1: The transmission control protocol (TCP) is a protocol in the TCP/IP suite that operates at the transport layer to provide connection-oriented, guaranteed delivery of packets. Hosts must first establish a session to exchange data and then confirm the delivery of packets using acknowledgments when using TCP. TCP is relatively slow in comparison to UDP. The hypertext transfer protocol secure (HTTPS) is a secure protocol used to provide web content to browsers using SSL/TLS encryption over TCP port 443. The simple network management protocol (SNMP) is a protocol for monitoring and managing network devices over UDP ports 161 and 162. A management system collates data sent by agents running on each device. The agents maintain a Management Information Base of configuration and usage data. An agent can also generate a trap, alerting the management system of some notable event (such as a printer that is out of paper). The trivial file transfer protocol (TFTP) is a protocol used to get a file from a remote host or put a file onto a remote host. TFTP is commonly used with embedded devices or systems that retrieve firmware, configuration information, or a system image during the boot process. TFTP operates over UDP port 69. The dynamic host control protocol (DHCP) is a protocol used to allocate IP addresses to a host when it joins a network. DHCP utilizes UDP ports 67 and 68.
For support or reporting issues, include Question ID: 63fe04ee3b7322449ddbaee0 in your ticket. Thank you.

4 0 / 1 point Multiple Choice

You want to play computer-based video games from anywhere in the world using your laptop or tablet. You heard about a new product called a Shadow PC that is a virtualized Windows 10 Home gaming PC in the cloud. Which of the following best describes this type of service?

- DaaS
- IaaS
- ✗ PaaS
- SaaS

Feedback

General Feedback

OBJ 4.1: Desktop as a Service (DaaS) provides a full virtualized desktop environment from within a cloud-based service. This is also known as VDI (Virtualized Desktop Infrastructure) and is coming in large enterprise businesses focused on increasing their security and minimizing their operational expenses. Shadow PC (shadow.tech) provides a version of DaaS for home users who want to have a gaming PC without all the upfront costs. Software as a Service (SaaS) is a cloud computing service that enables a service provider to make applications available over the Internet to end-users. This can be a calendar, scheduling, invoicing, word processor, database, or other programs. For example, Google Docs and Office 365 are both word processing SaaS solutions. Platform as a Service (PaaS) is a cloud computing service that enables consumers to rent fully configured systems that are set up for specific purposes. Infrastructure as a Service (IaaS) is a cloud computing service that enables a consumer to outsource computing equipment purchases and running their own data center.
For support or reporting issues, include Question ID: 63fe053a3b7322449ddbb287 in your ticket. Thank you.

5 1 / 1 point Multiple Choice

Christina recently purchased a new Android smartphone and is going on a trip. At the airport, she found a public wireless network called "FreeAirportWiFi" and connected to it. She noticed a question mark (?) icon showing in the toolbar next to the Wi-Fi icon. Christina attempts to open a webpage but gets an error of "The page cannot be displayed." She begins to troubleshoot the device by verifying that the airplane mode is disabled, Bluetooth is enabled, and tethering is enabled. Next, Christina attempts to make a phone call, which works without any issues. Which of the following is MOST likely the issue with Christina's smartphone?

- The smartphone does not have a valid data plan enabled
- ✓ The smartphone is connected to the FreeAirportWifi but is not authenticated yet

- ◯ The smartphone's SIM card is deactivated
- ◯ The smartphone can only support 3G data networks

Feedback

General Feedback

OBJ 5.5: When an Android smartphone is connected to the Wi-Fi but shows a question mark (?) next to the Wi-Fi's radio icon, this indicates that there is a lack of internet connectivity on the current wireless network. It appears that Christina's smartphone is fully connected to the FreeAirportWiFi, but she has not completed the authentication. These types of public wireless networks often have a captive portal or redirect page with the Acceptable Use Policy that must be accepted before giving the smartphone full connectivity to the internet. Once the acceptance is made to the captive portal, the smartphone is logically connected to the internet, and the question mark will be removed.
For support or reporting issues, include Question ID: 63fe055d3b7322449ddbb43f in your ticket. Thank you.

6 1 / 1 point Multiple Choice

Rick has just received a new Google Pixel smartphone and installed all of his favorite apps, including a different style virtual keyboard. After about 6 months of use, he noticed that the smartphone has become slow, and the virtual keyboard is not responding properly when keys are pressed or becomes completely unresponsive. What is the FIRST solution to try to fix this issue?

- ◯ Factory reset the phone
- ✓ ◉ Close the app in the background and restart the app
- ◯ Soft reset the phone
- ◯ Clear system cache from recovery mode

Feedback

General Feedback

OBJ 5.4: The simplest and most efficient way to fix unresponsive apps is to close the app running in the background and restart the app again. A custom virtual keyboard app can often cause performance decline and unresponsiveness, and the virtual keyboard will

lock up. It is always a good practice to always close apps running in the background to reduce resource consumption. If this doesn't solve the issue, then a soft reset or factory reset may be needed.</p>
For support or reporting issues, include Question ID: 63fe058a3b7322449ddbb660 in your ticket. Thank you.

7 1 / 1 point Multiple Choice

A user reports that their laptop's trackpad is unresponsive after waking the device from sleep mode. They have verified that the device works fine when using an external USB mouse. What should you check first to resolve the issue?

- ◯ Remove the external USB mouse and restart the laptop
- ◯ Update the laptop's BIOS to the latest version
- ◯ Reinstall the operating system to restore default trackpad functionality
- ✓ ◉ Confirm if the trackpad is disabled in the laptop's settings or function keys

Feedback

General Feedback

OBJ 1.2 - The first step is to confirm if the trackpad is disabled in the laptop's settings or function keys. Many laptops have a keyboard shortcut or a toggle in the system settings to enable or disable the trackpad, and it is possible the trackpad was accidentally turned off. Updating the BIOS or reinstalling the operating system are more advanced steps that should only be attempted if the settings check does not resolve the issue. Simply removing the USB mouse is unlikely to affect the trackpad's functionality.
For support or reporting issues, include Question ID: 6797da9c8e2e7a98bb10d763 in your ticket. Thank you.

8 1 / 1 point Multiple Choice

Which of the following types of networks would be created by installing an 802.11ax access point inside the Dion Training offices?

- ✓ ◉ WLAN

- WAN
- MAN
- SAN

Feedback

General Feedback

OBJ 2.7: A wireless LAN (WLAN) connects computers within a small and specific geographical area using the 802.11 protocols for their wireless connections. A storage area network (SAN) is a dedicated, independent, high-speed network that interconnects and delivers shared pools of storage devices to multiple servers. A storage area network often uses Fibre Channel over Ethernet (FCoE), Fibre Channel, or Internet Small Computer Systems Interface (iSCSI) connections to achieve the speeds necessary to support SAN operations. A metropolitan area network (MAN) is a computer network that interconnects users with computer resources in a geographic region of the size of a metropolitan or city area. A wide area network (WAN) is a network that spans multiple geographic locations, such as the internet. Wide area networks are often established with leased telecommunication circuits.
For support or reporting issues, include Question ID: 63fe04ca3b7322449ddbad2e in your ticket. Thank you.

9 0 / 1 point Multiple Choice

Dion Training has just installed a web server for a new domain name. Which of the following DNS records would need to be created to allow users to reach the website using its domain name and then redirect them to the proper IPv4 address for the server?

- SOA
- A
- ✕ MX
- AAAA

Feedback

General Feedback

OBJ 2.4: An A record associates your domain name with an IPv4 address. An AAAA

record associates your domain name with an IPv6 address. An MX record is used for outgoing (SMTP) and incoming (POP3/IMAP) traffic. A Start of Authority (SOA) resource record indicates which Domain Name Server (DNS) is the best source of information for the specified domain.
For support or reporting issues, include Question ID: 63fe04b43b7322449ddbac1d in your ticket. Thank you.

10 0 / 1 point Multiple Choice

Jason wants to order a new gaming laptop for his son for his birthday. The laptop display must support a high refresh rate of 120 Hz or 144 Hz. Which of the following types of laptop displays should Jason purchase to meet this requirement?

- ○ Plasma
- ○ VA
- ○ TN
- ✕ ◉ IPS

Feedback

General Feedback

OBJ 3.1 - A twisted nematic (TN) LCD panel uses technology where the panel is black when no electric current is running through the liquid crystal cells because the cells align themselves in a twisted state. When an electric current is applied, the liquid crystal cells untwist, allowing light to pass through, resulting in a white display screen. TN displays can be made with high refresh rates of 120 Hz or 144 Hz which makes them popular with gamers. An in-plane switching (IPS) LCD panel uses technology designed to resolve the quality issues inherent in TN panel technology, including strong viewing angle dependence and low-quality color reproduction. IPS displays have great color accuracy, but they do not support higher refresh rates sought after by gamers. A vertical alignment (VA) LCD panel uses technology designed to have liquid crystals lay perpendicular to the glass substrate when power is removed and in a vertical position when voltage is applied. This allows light to move through the liquid pixels when voltage is applied and the image is then produced. VA displays are a good middle-ground between IPS and TN panel displays. A plasma display is a type of flat panel display that uses small cells containing plasma with ionized gas that responds to electric fields. Plasma displays create excessive heat and are not as energy efficient as an LCD, LED, or OLED display.
For support or reporting issues, include Question ID: 63fe04973b7322449ddbaaba in your ticket. Thank you.

11 1 / 1 point Multiple Choice

After installing an SSD in a mobile device, the system does not detect the new drive. What is the most logical first step to resolve this issue?

- ◯ Update the firmware of the mobile device to the most recent version
- ✓ ◉ Check that the SSD is securely connected and properly seated in the device
- ◯ Format the drive using a file system that the mobile device supports
- ◯ Test the system with a different SSD to determine if the original is faulty

Feedback

General Feedback

OBJ 1.1 - The first step should be to ensure that the SSD is securely connected and properly seated. Connection issues are a common reason for hardware not being recognized. If the drive is correctly installed, further steps such as updating the firmware or formatting the drive can be considered. Testing another SSD should only be done after ruling out connection and configuration problems.
For support or reporting issues, include Question ID: 6797d7bbaad21b81318d3eea in your ticket. Thank you.

12 0 / 1 point Multiple Choice

Dion Training uses a VoIP conferencing solution to conduct its weekly staff meetings. When Jason is talking, some of the employees say it sounds like he is speeding up and slowing down randomly. Tamera is sitting in the office with Jason, and she says Jason is speaking at the same rate the entire time. Which of the following network performance metrics would be most useful in determining why the VoIP service is not presenting a consistent pace when delivering Jason's voice over the network?

- ◯ Bandwidth
- ◯ Jitter
- ✕ ◉ Latency
- ◯ Throughput

Feedback

General Feedback

OBJ 5.5: Jitter is a network condition that occurs when a time delay in the sending of data packets over a network connection occurs. Jitter is a big problem for any real-time applications you may be supporting on your networks, like video conferences, voice-over IP, and virtual desktop infrastructure clients. A jitter is simply a variation in the delay of the packets, and this can cause some strange side effects, especially for voice and video calls. If you have ever been in a video conference where someone was speaking and then their voice started speeding up for 5 or 10 seconds and then returned to normal speed, you have been on the receiving end of their network's jitter. Latency is the measure of time that it takes for data to reach its destination across a network. Usually, we measure network latency as the round-trip time from a workstation to the distant end and back. Throughput is an actual measure of how much data is successfully transferred from the source to a destination. Bandwidth is the maximum rate of data transfer across a given network. Now, bandwidth is more of a theoretical concept that measures how much data could be transferred from a source to a destination under ideal conditions. Therefore, we often measure throughput, instead of bandwidth, to monitor our network performance.

For support or reporting issues, include Question ID: 63fe05933b7322449ddbb6d3 in your ticket. Thank you.

13 1 / 1 point Multiple Choice

While working as a technician, you are trying to help a customer whose PC cannot boot up. When the computer is turned on, an error stating "bootable device not found" appears on the screen. What should you attempt first?

- ◯ Boot into Recovery Mode and run the Bootrec /fixmbr command to fix the MBR that's preventing the PC from booting
- ✓ ◉ Press the appropriate function key to check the boot order and select the proper HDD to boot from
- ◯ Verify that either the Molex or SATA cable is connected to the HDD
- ◯ Check Disk Management to see if the system drive is recognized

Feedback

General Feedback

OBJ 5.1: When the error "bootable device not found" appears, it typically indicates

that the system cannot find a drive with a valid operating system to boot from. The quickest and least invasive step is to check the boot order in the BIOS/UEFI or use a boot menu (often accessed by pressing a key like F12) to ensure the correct drive is selected as the boot device. Verifying that either the Molex or SATA cable is connected to the HDD involves physically opening the PC, which is more invasive and time-consuming than checking the boot order and may be necessary later if the drive is still not detected. However, it Booting into Recovery Mode and running the Bootrec /fixmbr command to fix the MBR assumes the issue is with the Master Boot Record (MBR), which may not be the case. Additionally, if the system cannot find a bootable device, you likely won't be able to access Recovery Mode. Disk Management is part of the operating system, and if the PC cannot boot, you won't be able to access it.</p>
For support or reporting issues, include Question ID: 63fe055e3b7322449ddbb444 in your ticket. Thank you.

14 0 / 1 point Multiple Choice

Which type of wireless network utilizes the 5 GHz frequency band and reaches speeds of up to 3.5 Gbps?

- 802.11a
- 802.11ac
- 802.11b
- 802.11g
- × 802.11n
- 802.11ax

Feedback

General Feedback

The 802.11ac (Wireless AC or Wi-Fi 5) standard utilizes a 5 GHz frequency to provide wireless networking at theoretical speeds up to 3.5 Gbps. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. The 802.11ax (Wireless AX or Wi-Fi 6) standard utilizes 2.4 GHz and 5.0 GHz frequencies to provide wireless networking at theoretical speeds up to 9.6 Gbps. Wireless AC uses orthogonal frequency-division multiple access (OFDMA) to conduct multiplexing of the frequencies transmitted and received to each client to provide additional bandwidth. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional

bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless AC also has a version called Wi-Fi 6E that supports the 6GHz frequency instead of the 2.4 GHz and 5.0 GHz frequencies used in Wi-Fi 6. The 802.11a (Wireless A) standard utilizes a 5 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11b (Wireless B) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 11 Mbps. The 802.11g (Wireless G) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11n (Wireless N) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 108 Mbps or a 5.0 GHz frequency to provide wireless networking at speeds up to 600 Mbps. Wireless N supports the use of multiple-input-multiple-output (MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless N supports channel bonding by combining two 20 MHz channels into a single 40 MHz channel to provide additional bandwidth.
For support or reporting issues, include Question ID: 63fe04f13b7322449ddbaf0d in your ticket. Thank you.

15 1 / 1 point Multiple Choice

Which button on a desktop computer begins a reboot of the computer without power being removed from the computer's components?

- ◯ Power
- ✓ ◉ Reset
- ◯ Hibernate
- ◯ Suspend

Feedback

General Feedback

OBJ 3.5: The reset button is usually located on the front panel of a desktop computer next to the power button. When the reset button is pressed, it allows the user to reboot the computer without releasing power from the system's components. This is often used when software has malfunctioned and caused the computer to halt or lock up. The power button is used to shut down or put the computer to sleep. There is usually not a dedicated suspend or hibernate button, but these are instead provided as software buttons in the operating system.
For support or reporting issues, include Question ID: 63fe052c3b7322449ddbb1de in your ticket. Thank you.

16 1 / 1 point Multiple Choice

Which of the following devices could be used to extend transmissions of a signal over longer distances by receiving and retransmitting a given signal at the lowest cost?

- ◯ Switch
- ✓ ◉ Repeater
- ◯ Firewall
- ◯ Router

Feedback

General Feedback

OBJ 2.5: A repeater is an electronic device that receives a signal and retransmits it. Repeaters are used to extend transmissions so that the signal can cover longer distances or be received on the other side of an obstruction. Repeaters may be wired or wireless, and they rebroadcast whatever signal is received through their transmitter. A firewall is a network security system that monitors and controls the incoming and outgoing network traffic based on predetermined security rules. A firewall typically establishes a barrier between a trusted internal network and an untrusted external network, such as the internet. A switch is a network device that receives incoming data into a buffer, then the destination MAC address is compared with an address table so that data is only sent out to the port with the corresponding MAC address. In a switched network, each port is in a separate collision domain. A router is a network device that links dissimilar networks and can support multiple alternate paths between locations based upon the parameters of speed, traffic loads, and cost. Routers are used to logically divide networks into subnets.
For support or reporting issues, include Question ID: 63fe04cc3b7322449ddbad42 in your ticket. Thank you.

17 1 / 1 point Multiple Choice

During the process of replacing an SSD in a mobile device, which precaution is most important to protect the component from damage?

- ◯ Ensure the mobile device is fully charged before beginning the replacement process
- ◯ Store the new SSD near other electronic devices to keep it readily accessible

✓ Wear an anti-static wrist strap and work on a surface that prevents electrostatic discharge

Use a magnetic screwdriver when handling screws near the drive

Feedback

General Feedback

OBJ 1.1 - Electrostatic discharge is a significant risk to sensitive components like SSDs. Wearing an anti-static wrist strap and working on an ESD-safe surface is the best practice to prevent damage during the installation. Magnetic screwdrivers can interfere with electronics and should not be used. Charging the device is unnecessary, and storing the SSD near other electronics can increase the risk of static buildup.
For support or reporting issues, include Question ID: 6797d765aad21b81318d3e8a in your ticket. Thank you.

18 0 / 1 point Multiple Choice

Which of the following protocols operates over port 139 by default?

✗ SMB

HTTPS

NetBIOS

IMAP

Feedback

General Feedback

<p>OBJ 2.1: The network basic input/output system (NetBIOS)/NetBIOS over TCP/IP (NetBT) is a session management protocol used to provide name registration and resolution services on legacy Microsoft networks and those using WINS. NetBIOS/NetBT operates on TCP/UDP ports 137 and 139. The internet message access protocol (IMAP) is a TCP/IP application protocol that provides a means for a client to access email messages stored in a mailbox on a remote server using TCP port number 143. Unlike POP3, messages persist on the server after the client has downloaded them. IMAP also supports mailbox management functions, such as creating subfolders and access to the same mailbox by more than one client at the same time. The hypertext transfer protocol secure (HTTPS) is a secure protocol used to provide web content to

browsers using SSL/TLS encryption over port 443. The server message block (SMB)/common internet file system (CIFS) is a protocol used for requesting files from Windows servers and delivering them to clients. SMB allows machines to share files and printers, thereby making them available for other machines to use. Samba software allows UNIX and Linux servers or NAS appliances to run SMB services for Windows clients. SMB operates on port 445.
For support or reporting issues, include Question ID: 63fe04c83b7322449ddbad15 in your ticket. Thank you.

19 1 / 1 point Multiple Choice

Which of the following devices would you configure an access control list (ACL) to implement a security policy within your network?

- ◯ Switch
- ◯ Hub
- ✓ ◉ Firewall
- ◯ Repeater

Feedback

General Feedback

OBJ 2.5: A firewall is a network security system that monitors and controls the incoming and outgoing network traffic based on predetermined security rules called an access control list (ACL). A firewall typically establishes a barrier between a trusted internal network and an untrusted external network, such as the internet. A switch is a network device that receives incoming data into a buffer, then the destination MAC address is compared with an address table so that data is only sent out to the port with the corresponding MAC address. In a switched network, each port is in a separate collision domain. A switch is a network device that receives incoming data into a buffer, then the destination MAC address is compared with an address table so that data is only sent out to the port with the corresponding MAC address. In a switched network, each port is in a separate collision domain. A hub provides no intelligence in its interconnection functions so that whatever is received on one port is repeated out every other port. This leads to many collisions occurring on the hub and increases the number of rebroadcasts, which can slow down the entire network.
For support or reporting issues, include Question ID: 63fe04b83b7322449ddbac4b in your ticket. Thank you.

20 1 / 1 point Multiple Choice

A user has reported that their workstation is running slowly. You perform a reboot of their workstation and receive a S.M.A.R.T. error during the boot-up process. Which of the following actions should you perform FIRST?

- Run diskpart
- Run chkdsk
- ✓ Backup the hard drive
- Reformat the hard drive

Feedback

General Feedback

OBJ 5.1: S.M.A.R.T. is an acronym for Self-Monitoring and Repair Tool. It is a feature in all modern magnetic hard drives (non-SSD drives) that monitors the hard drive to ensure it performs properly. S.M.A.R.T. can detect when the failure of a drive is imminent and can alert the user so that they can back up the drive before a complete failure occurs. If your hard drive produces a S.M.A.R.T. failure, you should immediately back up the drive. Once a backup has been completed, you can instead focus on repairing the drive using chkdsk.
For support or reporting issues, include Question ID: 63fe055c3b7322449ddbb430 in your ticket. Thank you.

21 1 / 1 point Multiple Choice

A network technician was tasked with installing a network printer and sharing it with a group of five instructors at Dion Training. The technician plugged the device into a switch port and noticed the link light turned green. Unfortunately, the printer was unable to obtain an IP address automatically. Which of the following is a potential reason for this error?

- Split-horizon is disabled
- Incorrect TCP port in ACL
- ✓ DHCP scope is exhausted
- Incorrect DNS records

Feedback

General Feedback

OBJ 2.4: The DHCP scope is used as a pool of IP addresses that can be assigned automatically. The issue might be that there are no more IP addresses left in the scope and it is therefore exhausted. The Dynamic Host Configuration Protocol (DHCP) uses port 67 and is a network management protocol used on Internet Protocol (IP) networks for automatically assigning IP addresses and other communication parameters to devices connected to the network using a client-server architecture. The Domain Name System (DNS) uses port 53 and is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network. DNS records are used to bind a domain name to an IP address using static assignments. Split horizon is a method used by distance vector protocols to prevent network routing loops. With split horizon, if a router receives routing information from another router, the first router will not broadcast that information back to the second router, thus preventing routing loops from occurring. An access control list (ACL) is a list of permissions associated with a system resource (object). Since the scenario specifies that the printer was unable to obtain an IP address automatically, it is most likely a DHCP issue.
For support or reporting issues, include Question ID: 63fe04ed3b7322449ddbaed1 in your ticket. Thank you.

22 0 / 1 point Multiple Choice

What ports do FTP and SFTP utilize?(This is a simulated Performance-Based Question. On the real exam, you may be given a chart with numerous ports and protocols and be asked to drag and drop them to match the ports with the protocols.)

- ○ 21, 23
- × ● 20, 21
- ○ 21, 22
- ○ 22, 23

Feedback

General Feedback

OBJ 2.1: FTP (File Transfer Protocol) uses ports 20 and 21. SFTP (Secure File Transfer Protocol) uses port 22. Port 23 is used by Telnet. If this were a question on the real exam, you would see a list of ports on one side and a list of protocols on the other,

and you would drag and drop each one to match them up. (It might also have 4-6 different pairs to match up.)
For support or reporting issues, include Question ID: 63fe04c43b7322449ddbace8 in your ticket. Thank you.

23 0 / 1 point Multiple Choice

You are currently conducting passive reconnaissance in preparation for an upcoming penetration test against Dion Training. You are reviewing the DNS records for the company and are trying to identify which third-party hosted services they may be using. Which of the following DNS records should you analyze to identify any human-readable records, domain verifications, and domain authentications

- SRV
- TXT
- MX
- × NS

Feedback

General Feedback

OBJ 2.4: Text (TXT) records are used to provide information about a resource, such as a server, network, or service, in human-readable form. They often contain domain verification and domain authentications for third-party tools that can send information on behalf of a domain name. Mail Exchange (MX) records are used to provide the mail server that accepts email messages for a particular domain. Nameserver (NS) records are used to list the authoritative DNS server for a particular domain. Service (SRV) records are used to provide host and port information on services such as voice over IP (VoIP) and instant messaging (IM) applications.
For support or reporting issues, include Question ID: 63fe04da3b7322449ddbadf3 in your ticket. Thank you.

24 0 / 1 point Multiple Choice

Which type of wireless network utilizes the 2.4 GHz frequency band and reaches speeds of up to 54 Mbps?

- ◯ 802.11b
- ✕ ◉ 802.11a
- ◯ 802.11n
- ◯ 802.11ac
- ◯ 802.11ax
- ◯ 802.11g

Feedback

General Feedback

<p>OBJ 2.2: The 802.11g (Wireless G) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11a (Wireless A) standard utilizes a 5 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11b (Wireless B) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 11 Mbps. The 802.11n (Wireless N) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 108 Mbps or a 5.0 GHz frequency to provide wireless networking at speeds up to 600 Mbps. Wireless N supports the use of multiple-input-multiple-output (MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless N supports channel bonding by combining two 20 MHz channels into a single 40 MHz channel to provide additional bandwidth. The 802.11ac (Wireless AC or Wi-Fi 5) standard utilizes a 5 GHz frequency to provide wireless networking at theoretical speeds up to 3.5 Gbps. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. The 802.11ax (Wireless AX or Wi-Fi 6) standard utilizes 2.4 GHz and 5.0 GHz frequencies to provide wireless networking at theoretical speeds up to 9.6 Gbps. Wireless AX uses orthogonal frequency-division multiple access (OFDMA) to conduct multiplexing of the frequencies transmitted and received to each client to provide additional bandwidth. Wireless AX uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AX uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless AX also has a version called Wi-Fi 6E that supports the 6 6GHz frequency instead of the 2.4 GHz and 5.0 GHz frequencies used in Wi-Fi 6. The 802.11g (Wireless G) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11a (Wireless A) standard utilizes a 5 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11b (Wireless B) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 11 Mbps. The 802.11n (Wireless N) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 108 Mbps or a 5.0 GHz frequency to provide wireless networking at speeds up to 600 Mbps. Wireless N supports the use of multiple-input-

multiple-output (MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless N supports channel bonding by combining two 20 MHz channels into a single 40 MHz channel to provide additional bandwidth. The 802.11ac (Wireless AC or Wi-Fi 5) standard utilizes a 5 GHz frequency to provide wireless networking at theoretical speeds up to 3.5 Gbps. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. The 802.11ax (Wireless AX or Wi-Fi 6) standard utilizes 2.4 GHz and 5.0 GHz frequencies to provide wireless networking at theoretical speeds up to 9.6 Gbps. Wireless AC uses orthogonal frequency-division multiple access (OFDMA) to conduct multiplexing of the frequencies transmitted and received to each client to provide additional bandwidth. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless AC also has a version called Wi-Fi 6E that supports the 6 6GHz frequency instead of the 2.4 GHz and 5.0 GHz frequencies used in Wi-Fi 6.
For support or reporting issues, include Question ID: 63fe04d33b7322449ddbad9c in your ticket. Thank you.

25 1 / 1 point Multiple Choice

A customer is trying to configure an 802.11b wireless card in an old laptop to connect to an 802.11g wireless router. When the customer scans for the wireless network's SSID (Dion-Corp), it is not displayed within Windows. What is the MOST likely reason that the SSID is not being displayed?

- ◯ 802.11g and 802.11b use different frequencies
- ◯ The wireless router is not configured for DHCP support
- ◯ The wireless router is configured with WPA2 encryption
- ✓ ◉ The broadcast is disabled on the wireless router

Feedback

General Feedback

OBJ 5.5: If the SSID (Secure Set Identifier) is disabled, then the wireless network name will not be broadcast to any available devices within range. Both Wireless B and G use the same frequency band (2.4 GHz) and would not cause this issue. Similarly, encryption that is enabled or disabled would not affect the SSID broadcast since the SSID is sent out in cleartext. DHCP support is used once a device connects to the network.

Therefore, it would not affect the SSID broadcast.
For support or reporting issues, include Question ID: 63fe058c3b7322449ddbb674 in your ticket. Thank you.

26 0 / 1 point Multiple Choice

Dion Training wants to order some new laptops for their video editors. The video editors want to ensure the display has the best color accuracy when using a 60 Hz refresh rate. Which of the following types of displays should the company purchase to meet this requirement?

- IPS
- × TN
- Plasma
- VA

Feedback

General Feedback

OBJ 3.1: An in-plane switching (IPS) LCD panel uses technology designed to resolve the quality issues inherent in TN panel technology, including strong viewing angle dependence and low-quality color reproduction. IPS displays have great color accuracy, but they do not support higher refresh rates sought after by gamers. A twisted nematic (TN) LCD panel uses technology where the panel is black when no electric current is running through the liquid crystal cells because the cells align themselves in a twisted state. When an electric current is applied, the liquid crystal cells untwist, allowing light to pass through, resulting in a white display screen. TN displays can be made with high refresh rates of 120 Hz or 144 Hz which makes them popular with gamers. A vertical alignment (VA) LCD panel uses technology designed to have liquid crystals lay perpendicular to the glass substrate when power is removed and in a vertical position when voltage is applied. This allows light to move through the liquid pixels when voltage is applied and the image is then produced. VA displays are a good middle-ground between IPS and TN panel displays. A plasma display is a type of flat panel display that uses small cells containing plasma with ionized gas that responds to electric fields. Plasma displays create excessive heat and are not as energy efficient as an LCD, LED, or OLED display.
For support or reporting issues, include Question ID: 63fe048d3b7322449ddbaa38 in your ticket. Thank you.

27 1 / 1 point Multiple Choice

Which of the following parts is responsible for converting the analog signals created by your tapping or swiping the surface of a display to a binary signal that is understood by the underlying software?

- Capacitor
- ✓ Digitizer
- Backlight
- Inverter

Feedback

General Feedback

OBJ 3.1 - The digitizer is a touch-sensitive glass panel covering the LCD in a touchscreen assembly. The digitizer is used to convert touch events to digital signals that can be interpreted as different types of input. The digitizer can be touched and inspected without taking apart the device. Analog signals are created when you tap or swipe the surface of the display. The digitizer is connected to the laptop with a flexible digitizer cable. A grid of sensors is activated when you tap or swipe the screen. The sensors' information is sent through the digitizer cable to a circuit that converts the analog signal to a digital signal using binary. A backlight is a fluorescent lamp used to illuminate the image on a flat panel (LCD) screen. If the backlight or inverter fails, the screen image will go very, very dark. The inverter (also known as inverter board, screen inverter, or LCD inverter) is a part of the laptop's display assembly. The inverter is a small green board attached to the base of the LCD panel. In a laptop computer, the LCD screen relies on a backlight source to uniformly light up the screen, and the inverter acts as the power supply for this lighting source. If the inverter is faulty, the display will be too dim to easily see without an external light source like a flashlight. A capacitor is an electrical component that stores electrical energy and is often used to regulate voltages. It can hold a charge after the power is removed.
For support or reporting issues, include Question ID: 63fe04953b7322449ddbaaa1 in your ticket. Thank you.

28 1 / 1 point Multiple Choice

An employee's laptop has become warm to the touch and does not sit evenly on the table during use. You turn the laptop over and see the case bulging into a slightly rounded shape. What is the MOST likely cause of this issue?

- Wireless card is defective
- Keyboard is defective
- ✓ Battery is defective
- RAM is defective

Feedback

General Feedback

Based on the description, the battery is defective and swelling. Swollen batteries, while not common, are a significant risk. They result from too much current inside a cell of the battery, which causes a build-up of heat and gas. This can be caused by overcharging, manufacturer defects, deep discharge, or damage to the battery. The most common cause is an overcharge of the battery, which causes a chemical reaction between the electrodes and the electrolyte, resulting in the release of heat and gases that expand inside the battery, causing the casing to swell or split open. If a battery begins to bulge, swell, or split, it should immediately be replaced.
For support or reporting issues, include Question ID: 63fe05893b7322449ddbb656 in your ticket. Thank you.

29 1 / 1 point Multiple Choice

There is a crinkling sound coming from the laser printer in the Dion Training offices. During their investigation, a technician observes that the images are printing at an angle on the pages. Which of the following should the technician do to fix this problem?

- ✓ Replace the rollers
- Replace the toner
- Clean the transfer drum
- Clear the page counter

Feedback

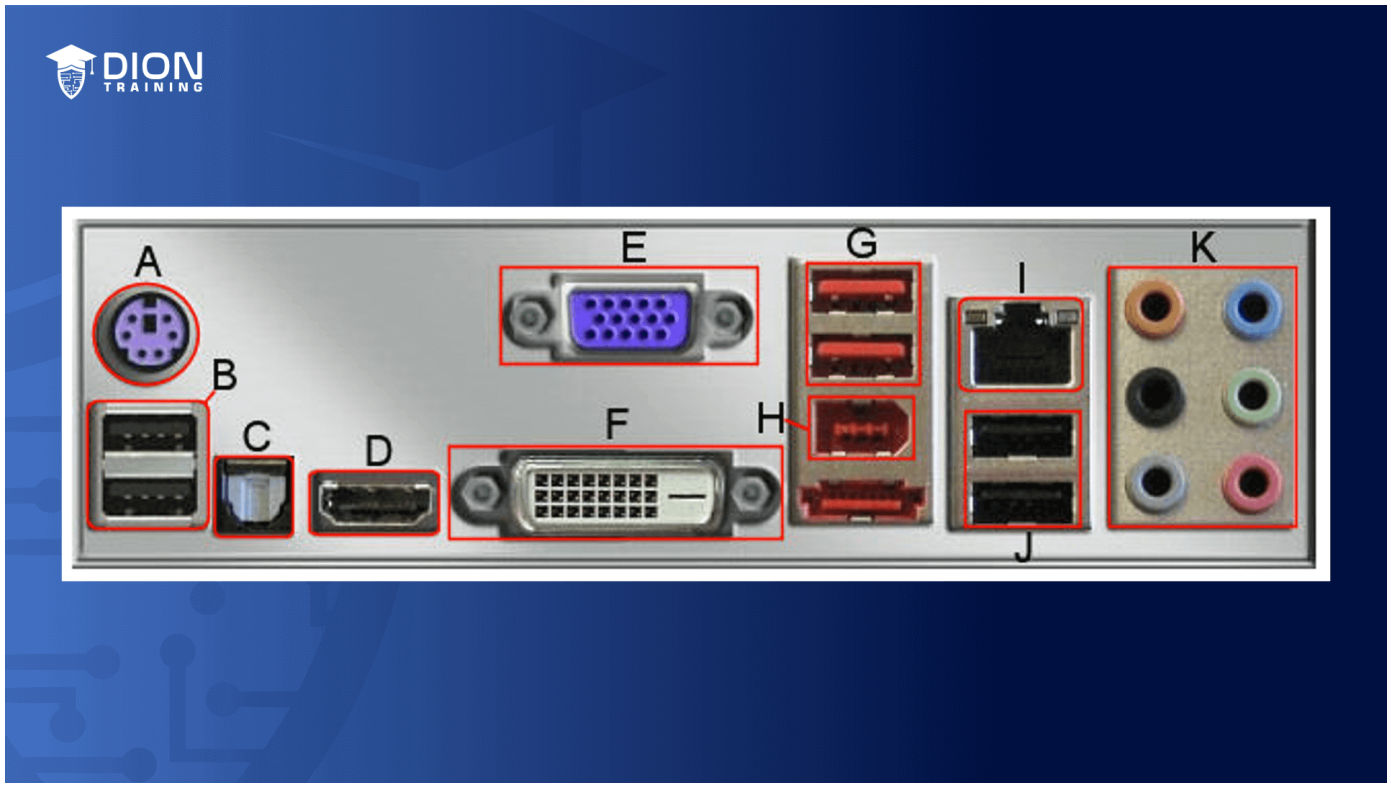
General Feedback

<p>OBJ 5.6: If the rollers become unevenly worn, they will feed the pages in at an angle,

causing the misalignment in the printed output. To fix this issue, the rollers should be replaced. The page, country, toner, and transfer drum will not affect the angle of the printed images on the paper.</p>
For support or reporting issues, include Question ID: 63fe05a63b7322449ddbb7b4 in your ticket. Thank you.

30 0 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as F?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- ✕ ◉ VGA
- ○ DVI-I
- ○ HDMI
- ○ DVI-D

Feedback

General Feedback

OBJ 3.2 - This port is the DVI-D connection. This is not a DVI-I since there is only a dash and not a cross on the right side of the port. This means it only supports digital and not analog connections. The digital video interface (DVI) came in three models: DVI-A (analog), DVI-D (digital), and DVI-D (analog and digital known as integrated). DVI adapters were designed to replace the VGA ports used in legacy CRT monitors but they were themselves replaced by HDMI. While this is an older technology, you will still see them commonly used in corporate environments. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.
For support or reporting issues, include Question ID: 63fe05333b7322449ddbb233 in your ticket. Thank you.

31 1 / 1 point Multiple Choice

You are troubleshooting a workstation with four identical memory sticks installed, one in each of the slots labeled 1, 2, 3, and 4. The motherboard has a marking near the memory slots that says, "quad-channel support." Based on what you observe, how many channels are being used by the motherboard's memory?

- ○ Three channels
- ✓ ◉ Four channels
- ○ Two channels
- ○ One channel

Feedback

General Feedback

OBJ 3.3: Since the motherboard says it supports quad-channel and four sticks of memory are installed, it would be using four channels. By using a motherboard that supports a quad-channel memory controller, the motherboard is able to transfer up to 256 bits of memory per transaction (4 x 64 bits) to increase the bandwidth available for data transfer.
For support or reporting issues, include Question ID: 63fe05343b7322449ddbb23d in your ticket. Thank you.

32 1 / 1 point Multiple Choice

Dion Training recently hired a new graphic designer. You have been asked to install a new, high-end PCIe x16 graphics card into one of the standard corporate workstations so the new designer can get to work. After installing the card, you attempt to power on the computer. The computer makes some odd noises when you press the power button, but the monitor displays the words, 'no source.' What is MOST likely the issue?

- ◯ The workstation is overheating due to the graphics card
- ◯ The processor in the workstation isn't powerful enough to support the card
- ◯ The graphics card installed in the workstation was defective
- ✓ ◉ The workstation's power supply isn't powerful enough to support the card

Feedback

General Feedback

OBJ 5.1: High-end graphics cards generally require additional power to operate. Most corporate workstations come with minimal power supplies (usually 250-300 watts) that cannot provide the additional power required to operate a high-end graphics card. In this case, the power supply should be upgraded to at least a 500-750 watt power supply to support the new graphics card.
For support or reporting issues, include Question ID: 63fe058e3b7322449ddbb692 in your ticket. Thank you.

33 1 / 1 point Multiple Choice

Your friend just noticed their smartphone's battery is running low. Unfortunately, your friend isn't very knowledgeable about computers and smartphones, so they don't know what type of charging cable they need. They know their phone is running iOS 15 and state they obtained their phone back in 2020. What kind of cable does your friend need to charge their phone?

- ✓ ◉ Lightning
- ◯ USB-C
- ◯ Micro USB
- ◯ Mini USB

Feedback

General Feedback

OBJ 3.2 - Lightning cables are used by Apple's iPhone, which are the only smartphones that run the iOS software. Since your friend is using iOS, he uses an iPhone (smartphone) or an older iPad (tablet). Lightning is a proprietary computer bus and power connector created by Apple and is used by every iPhone from the 5 to 13 series. Since 2023, newer Apple devices have shifted to the use of USB-C for charging instead of the lightning cable, though. USB-C is the default standard used in USB 3.1 with a 24-pin connector. The connector is reversible and can be inserted in either direction. Micro USB is a miniaturized version of the Universal Serial Bus (USB) interface developed for connecting compact and mobile devices such as smartphones, Mp3 players, GPS devices, photo printers, and digital cameras. Micro USB connectors exist or have existed in three forms: micro A, micro B, and micro USB 3. Mini USB was introduced with USB 2.0 and was used with digital cameras and early tablets. Mini USB is not commonly used in modern devices.
For support or reporting issues, include Question ID: 63fe04a93b7322449ddbab96 in your ticket. Thank you.

34 0 / 1 point Multiple Choice

You have been asked to run to the store and purchase a network cable for your boss's new laptop. They need a network cable that supports at least 750 Mbps. Which of the following network cables should you purchase to meet the bandwidth required at the lowest price?

- ○ CAT 5
- ✕ ◉ CAT 6
- ○ CAT 7
- ○ CAT 5e

Feedback

General Feedback

OBJ 3.2 - You should purchase a CAT 5e cable since it can support up to 1000 Mbps (1 Gbps) of bandwidth and is the cheapest option provided that will meet your boss' bandwidth requirements. CAT 5 cables only support 100 Mbps. Therefore, it won't meet the requirements. CAT 6 and CAT 7 cables will support 1 Gbps and 10 Gbps, respectively, but they are more expensive than CAT 5e cables. Therefore, the CAT 5e cable is the best choice for the requirements provided.

For support or reporting issues, include Question ID: 63fe05043b7322449ddbaff3 in your ticket. Thank you.

35 1 / 1 point Multiple Choice

A company relies on legacy accounting software that is no longer compatible with the latest operating systems. Which of the following is the BEST solution to ensure continued use of the software without upgrading the operating system?

- ◯ Install the software on a cloud-based server
- ◯ Rewrite the application to work on the new OS
- ✓ ◉ Install a virtual machine running the legacy OS
- ◯ Use compatibility mode on the new OS

Feedback

General Feedback

OBJ 4.1: Installing a virtual machine running the legacy OS is the best solution because it allows the application to run in an isolated environment, ensuring compatibility while maintaining the existing infrastructure without impacting the host system. Using compatibility mode on the new OS is not reliable for complex legacy applications. Rewriting the application is costly and time-consuming. Installing it on a cloud-based server may not provide the required environment for legacy software.
For support or reporting issues, include Question ID: 67973dfc1fa68cb9f17d1bd1 in your ticket. Thank you.

36 0 / 1 point Multiple Choice

A penetration tester wants to build a workstation that will be used to brute-force hash digests. Which of the following is the BEST option to ensure sufficient power and speed to crack them?

- ✗ ◉ Multi-core CPU
- ◯ Integrated GPU

- Dedicated GPU
- 7200-RPM HDD

Feedback

General Feedback

OBJ 3.5: Dedicated GPUs are designed to conduct complex mathematical functions extremely quickly. If you want to build a system to perform the cracking of a password, hash, or encryption algorithm, it is important to have a high-speed, dedicated GPU. The reason to use a GPU instead of a CPU for password cracking is that it is much faster for this mathematically intensive type of work. Cracking passwords, hashes, and encryption is a lot like mining cryptocurrency in that using dedicated GPUs will give you the best performance.
For support or reporting issues, include Question ID: 63fe05083b7322449ddbb025 in your ticket. Thank you.

37 0 / 1 point Multiple Choice

A user wants to print 2 copies of a 3-page report at once. The user would like the pages to come out of the printer in the order of pages 1, 2, 3, 1, 2, 3 instead of pages 1, 1, 2, 2, 3, 3. What printer setting should the user enable for this print job to achieve this result?

- Transparency
- Orientation
- × Duplex
- Collate

Feedback

General Feedback

OBJ 3.7: The collate setting allows the gathering and arranging of individual sheets or other printed components into a predetermined sequence. Collating creates consistent, logical sets from multiple parts, such as printing a series of pages in a report like pages 1, 2, and 3 before printing the second copy. This makes it easier to staple and distribute to people after coming out of the printer. Duplex printing is a feature of some computer printers, multi-function printers (MFPs), and copy machines that allows the printing of a sheet of paper on both sides automatically. Without this capability, print

devices can only print on a single side of the paper, sometimes called single-sided printing or simplex printing. With duplex printing, information is printed on the front side of a piece of paper, and then the paper is automatically fed back through the printer to print the information on the backside of the paper. Page orientation is how a rectangular page is oriented for normal viewing. The two most common types of orientation are portrait and landscape. The term transparency was made up as a distractor for this question.
For support or reporting issues, include Question ID: 63fe05293b7322449ddbb1b1 in your ticket. Thank you.

38 1 / 1 point Multiple Choice

Which of the following cloud deployment models is dedicated to a single organization and offers enhanced security and control?

- ○ Public cloud
- ✓ ◉ Private cloud
- ○ Hybrid cloud
- ○ Community cloud

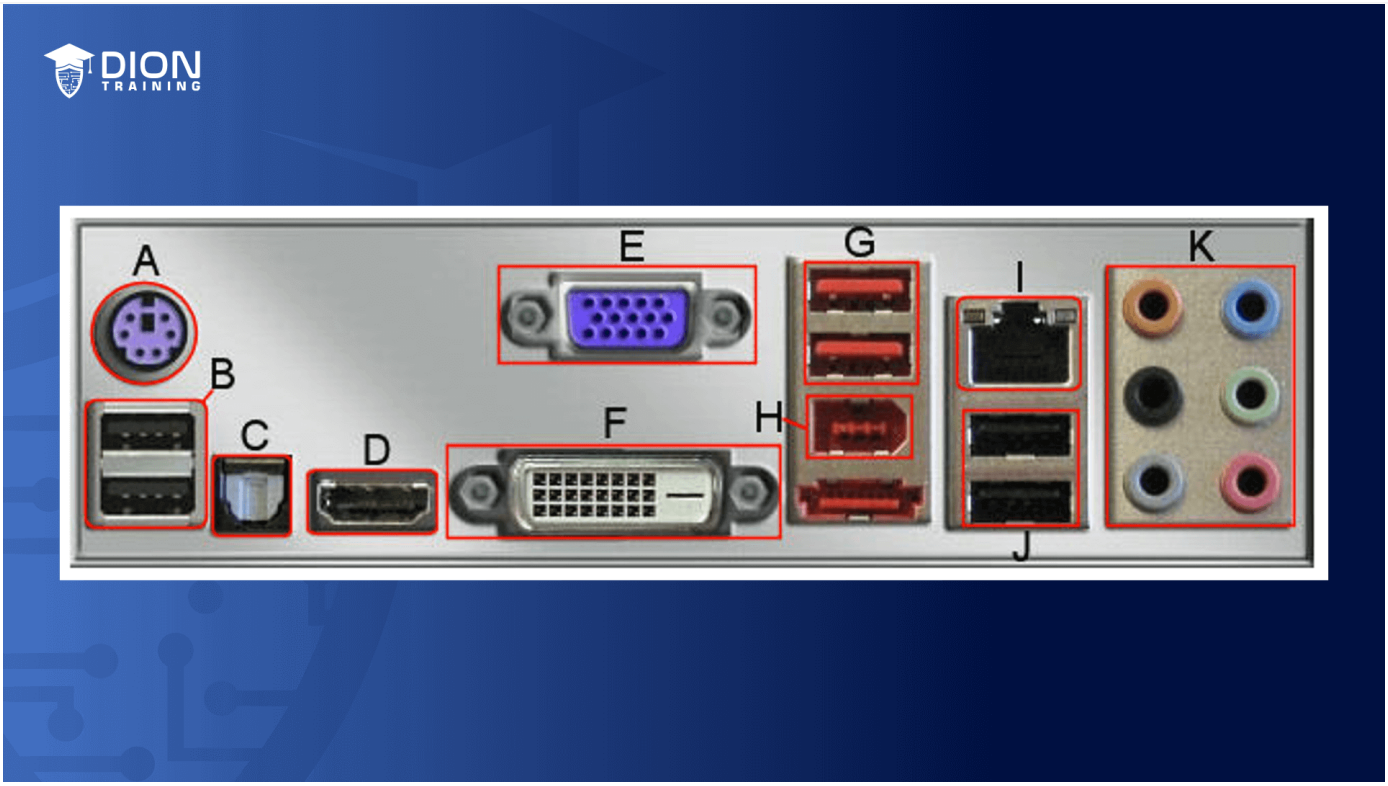
Feedback

General Feedback

OBJ 4.2: A private cloud is dedicated to a single organization, offering greater security, control, and compliance compared to public cloud environments. A public cloud provides resources to multiple organizations and individuals, reducing control and security. A community cloud is shared among multiple organizations with common interests or regulatory requirements. Hybrid clouds combine elements of both public and private clouds but do not provide the same level of dedicated control as a private cloud.
For support or reporting issues, include Question ID: 6797350da7a62dd7fe063c84 in your ticket. Thank you.

39 0 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as G?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- ✕ USB 2.0
- FireWire
- USB-C
- USB 3.0

Feedback

General Feedback

OBJ 3.2 - These ports are the USB 3.0 connections. The red coloring notates this. USB 3.0 ports are always shown as a different color to differentiate them from the older and slower USB 2.0 ports. Universal serial bus (USB) is the main type of connection interface used on desktops and laptops. A larger Type A connector attaches to a port on the host; Type B and Mini- or Micro-Type B connectors are used for devices. USB 1.1 supports 12 Mbps while USB 2.0 supports 480 Mbps and is backward compatible with 1.1 devices which run at a slower speed. USB devices are considered hot-swappable and each device can draw up to 2.5 W of power over a USB port. USB 3.0 and 3.1 define 5 Gbps (SuperSpeed) and 10 Gbps (SuperSpeed+) rates and can deliver up to 4.5 W of power. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.

For support or reporting issues, include Question ID: 63fe05223b7322449ddbb161 in your ticket. Thank you.

40 1 / 1 point Multiple Choice

An employee's workstation will operate for a short period of time, but then it unexpectedly shuts down. When you check the workstation, you hear the hum of the cooling fans. What is most likely the issue?

- [x] CPU overheated
- [ ] GPU overheated
- [ ] Defective case fan
- [ ] Defective NIC card

Feedback

General Feedback

OBJ 5.1: If a computer's CPU becomes overheated, the system will unexpectedly shut down the computer to protect the processor from damage. This usually occurs if the CPU's cooling fan isn't working properly, the heat sink is dislodged, or the thermal paste breaks down. In this case, you would need to remove the heat sink, remove the processor, reseat the processor, reapply thermal paste, reattach the heat sink, and reconnect the processor fan before rebooting the computer to solve this issue.
For support or reporting issues, include Question ID: 63fe059c3b7322449ddbb741 in your ticket. Thank you.

41 1 / 1 point Multiple Choice

Dion Training has recently purchased a new wireless printer and has asked you to configure it in their office. You have installed the necessary printer drivers on your workstation and have plugged the printer into the power outlet. You just configured the printer to connect to the guest wireless network. Next, you try to print a test page, but nothing prints out. Which of the following is the most likely cause for the printer not printing the test page?

- [ ] You didn't enable DHCP on your workstation

- [x] You connected the printer to the guest wireless network instead of the corporate wireless network
- [ ] You didn't configure the printer to use the Internet Printing Protocol (IPP)
- [ ] You forgot to configure your workstation as a print server

Feedback

General Feedback

OBJ 5.6: Most corporate wireless networks are configured as two separate networks: one for the corporate wireless network traffic and another for any guests that might connect to the wireless network while visiting your offices. These two wireless networks are isolated from each other, so if you connect the printer to the guest wireless network, then the workstations will be unable to reach the printer and print out any documents or test pages.
For support or reporting issues, include Question ID: 63fe056d3b7322449ddbb502 in your ticket. Thank you.

42 0 / 1 point Multiple Choice

Dion Training's graphic designer is having an issue printing full-page color images using an older color laser printer on the network. The graphic designer can print smaller images such as full-page vector graphic logos, but if they attempt to print a full-page bitmap photo, the printer will print only 60% to 70% of the page and then fail. The vector images are less than 100 MB in size, but the larger bitmap photos can be 400 MB to 800 MB in size. Which of the following is the MOST likely reason for the print failure of the photos?

- [x] The graphic designer is not printing using PCL
- [ ] The connection to the print server is timing out
- [ ] The printer is low on black toner
- [ ] The printer is running out of memory

Feedback

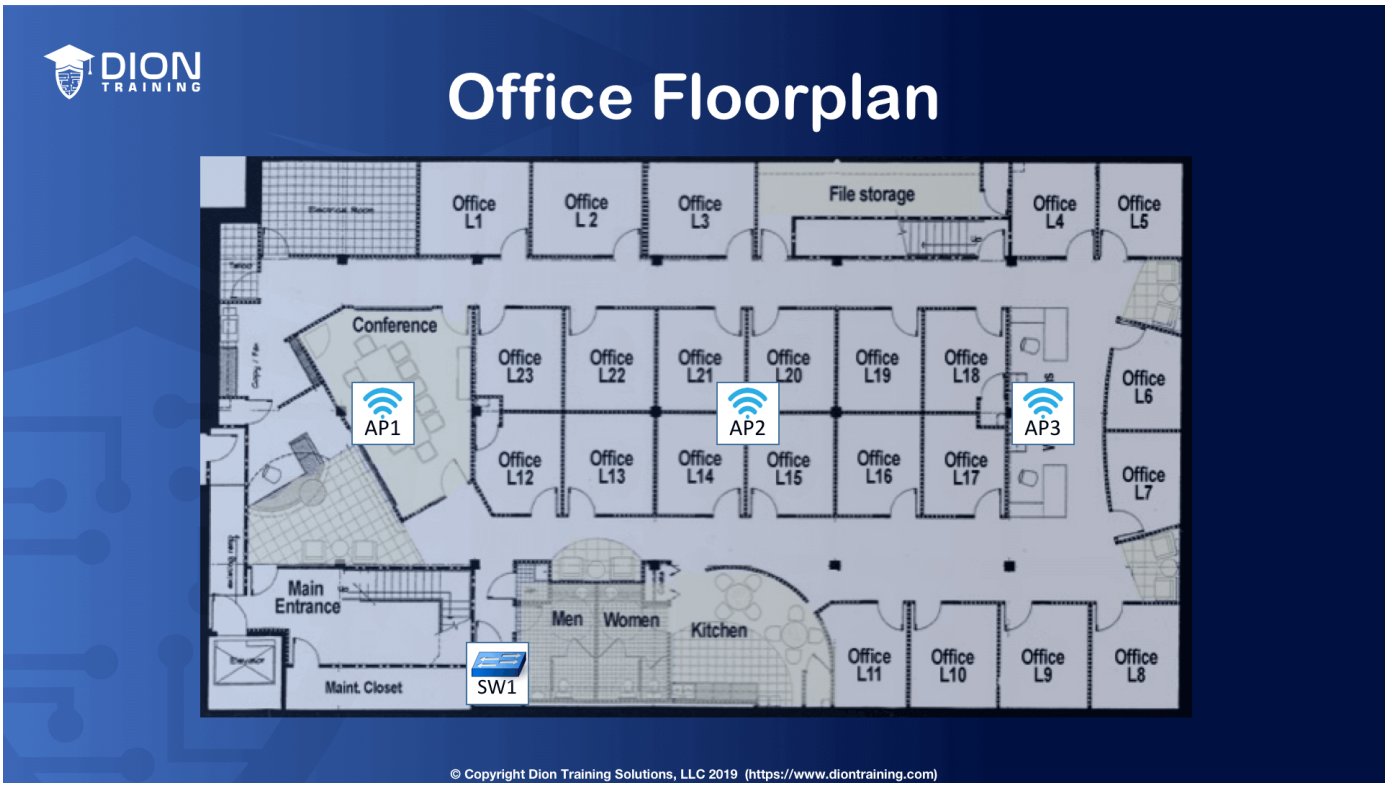
General Feedback

OBJ 5.6: The most likely reason for this failure is a lack of memory in the printer. A laser printer must load the entire image into memory before it can process it and print

the image to the page. If the laser printer runs out of memory, it will print the part of the image it has processed in memory to the page and then fail. ;For example, if this printer only has 256 MB of RAM, then it will be able to print the vector graphics without any issues and will fail when trying to print full-size high-resolution photos that are 400 MB to 800 MB in size. There is no indication that there is fading in the printed images to indicate the toner is low. Whether using PCL or PostScript, the lack of memory will still cause the print job to fail. The print server is not timing out since the printer can load most of the image to its memory before failing.</p>
For support or reporting issues, include Question ID: 63fe05973b7322449ddbb700 in your ticket. Thank you.

43 1 / 1 point Multiple Choice

Wireless network users recently began experiencing speed and performance issues on your network after Access Point 2 (AP2) was replaced due to a recent hardware failure. The original wireless network was installed according to a wireless consultant's specifications and has always worked properly without any past issues.You have been asked to evaluate the situation and resolve any issues you find to improve the network's performance and connectivity. The client has instructed you to adjust the least amount of settings/configurations possible while attempting to fix the issue. Before arriving on-site, you receive the below office's floor plan with an elementary network diagram drawn on top.Based on the information provided to you so far, which network device would you log into first to begin your troubleshooting efforts?



(This is a simulated Performance-Based Question. On the real certification exam, you will be asked to click on the appropriate device in a network diagram to see and modify its configuration.)

- AP1
- AP3
- ✓ AP2
- SW1

Feedback

General Feedback

OBJ 5.5: Since everything was working properly on the network before AP2 was replaced after the recent hardware failure, AP2 likely has some configuration error that has led to the recent connectivity and performance problems. Therefore, you should begin your troubleshooting efforts with AP2. According to the CompTIA troubleshooting method, you should always determine if anything has changed and question the obvious. If AP2 was recently replaced, it is most likely the device with an incorrect configuration setting or the one causing the issues.
For support or reporting issues, include Question ID: 63fe05763b7322449ddbb56b in your ticket. Thank you.

44 1 / 1 point Multiple Choice

Which of the following best describes the purpose of containers in virtualization?

- They emulate an entire physical machine, lessening the need for hardware
- They replace the need for operating systems
- ✓ They provide a lightweight, consistent runtime environment
- They eliminate the need for hypervisors

Feedback

General Feedback

OBJ 4.1: The best description of the purpose of containers in virtualization is that they provide a lightweight, consistent runtime environment because containers package applications with their dependencies, ensuring consistency across different

environments while using fewer resources than traditional virtual machines. Containers share the host OS kernel rather than virtualizing hardware; they do not replace the need for a physical machine. The use of operating systems is still required since containers still rely on an OS, although they do not require a full guest OS. While containers can run directly on the host OS, hypervisors are still needed for full virtual machine environments.
For support or reporting issues, include Question ID: 6797377ba7a62dd7fe063c9a in your ticket. Thank you.

45 1 / 1 point Multiple Choice

Dion Training's multifunction laser printer recently had a paper jam. An employee pulled the paper out of the rollers to clear the jam. Unfortunately, the device still has a paper jam error displayed on the screen and will not print. You have been called to fix this issue. Which of the following should you attempt FIRST?

- ◯ Replace the pickup rollers
- ◯ Update the drivers on the print server
- ◯ Replace the toner cartridge
- ✓ ◉ Verify the paper path and its sensors are not blocked

Feedback

General Feedback

OBJ 5.6: Often, employees will pull the jammed paper out of a printer to clear a paper jam. This can lead to small bits of paper being left in the path or in front of a blockage sensor. As a technician, you should always check the entire path is clear and the sensors are not blocked first if the printer claims there is a paper jam, as this is the most likely cause of a continued paper jam error.
For support or reporting issues, include Question ID: 63fe05a13b7322449ddbb778 in your ticket. Thank you.

46 1 / 1 point Multiple Choice

Which security measure is essential for protecting virtual machines from unauthorized access within a virtualization environment?

- Enabling automatic backups
- Using dynamic resource allocation
- ✓ Implementing RBAC
- Increasing virtual CPU allocation

Feedback

General Feedback

OBJ 4.1: Role-based access control (RBAC) restricts access to virtual machines based on the user's role within the organization, ensuring only authorized personnel can modify or access resources. Increasing virtual CPU (vCPU) allocation enhances performance, not security. Enabling automatic backups provides data protection but does not control access. Using dynamic resource allocation optimizes performance, not access control.

For support or reporting issues, include Question ID: 67973868a7a62dd7fe063ca9 in your ticket. Thank you.

47 0 / 1 point Multiple Choice

You are troubleshooting a customer's printer that is attached to their workstation by a USB cable. The printer is powered on, but it fails to print. You question the customer, but they claim that nothing has changed on the computer or the printer since it was printing earlier today. Which of the following should you do NEXT to determine the root cause of the printer failure?

- ✗ Unplug and reattach the printer cables and reboot the workstation
- Replace the printer with a new model and dispose of the broken printer
- Update the print drivers on the Windows workstation
- Examine the workstation and printer for any obvious indicators of failure

Feedback

General Feedback

OBJ 5.6: You should always question the obvious when establishing the theory of probable cause. In this case, you have not fully identified the problem yet, so you must establish a theory and test it. For the exam, it is important that you can list and identify the 6 steps of the CompTIA troubleshooting methodology in order. (1) Identify the problem. (2) Establish a theory of probable cause (question the obvious). (3) Test the theory to determine the cause. (4) Establish a plan of action to resolve the problem and then implement the solution. (5) Verify full system functionality and, if applicable, implement preventative measures. (6) Document findings, actions, and outcomes.
For support or reporting issues, include Question ID: 63fe05743b7322449ddbb54d in your ticket. Thank you.

48 0 / 1 point Multiple Choice

A company is transitioning to a cloud-based infrastructure and wants to implement a pay-as-you-go model for its IT services. Which cloud deployment model should they choose?

- ◯ Private cloud
- ✕ ◉ Community cloud
- ◯ Hybrid cloud
- ◯ Public cloud

Feedback

General Feedback

OBJ 4.2: The public cloud model allows organizations to pay only for the resources they use, making it a cost-effective solution for businesses that want to scale their IT services without heavy upfront investments. Public cloud providers offer shared infrastructure, which reduces costs but may have security and compliance considerations.
For support or reporting issues, include Question ID: 67973c0622cbe20a3b2640b2 in your ticket. Thank you.

49 0 / 1 point Multiple Choice

Which of the following is a proprietary wireless communication protocol operating in the 800 MHz to 900 MHz frequency band and is primarily used for home automation?

- NFC
- ✗ Zigbee
- Z-Wave
- Bluetooth

Feedback

General Feedback

OBJ 2.2: Z-Wave is a wireless communications protocol used primarily for home automation. Z-Wave creates a mesh network topology, using low-energy radio waves to communicate from one appliance to another. Devices can be configured to work as repeaters to extend the network, but there is a limit of four hops between a controller device and an endpoint. This allows for wireless control of residential appliances and other devices, such as lighting control, security systems, thermostats, windows, locks, swimming pools, and garage door openers. Zigbee is a low-power wireless communications open-source protocol used primarily for home automation. Zigbee uses radio frequencies in the 2.4 GHz band and a mesh topology. Near-field communication (NFC) is a set of communication protocols that enable two electronic devices, one of which is usually a portable device such as a smartphone, to establish communication by bringing them within 4 cm of each other. This is commonly used for contactless payment systems, transferring contacts, or transferring a file from one device to another. Bluetooth is a wireless technology standard used for exchanging data between fixed and mobile devices over short distances using UHF radio waves in the industrial, scientific, and medical radio bands from 2.402 GHz to 2.480 GHz and building a personal area network (PAN). Bluetooth is commonly used when connecting wireless devices like mice, trackpads, headphones, and other devices.
For support or reporting issues, include Question ID: 63fe04c03b7322449ddbacb1 in your ticket. Thank you.

50 1 / 1 point Multiple Choice

Which of the following types of connectors are used to terminate a household telephone cable?

- ✓ RJ-11
- LC

- F type
- RJ-45

Feedback

General Feedback

OBJ 3.2 - An RJ-11 (Registered Jack 11) is a telephone interface that uses a cable of twisted wire pairs and a modular jack with two, four, or six contacts. RJ-11 is the common connector for plugging a household telephone into the wall and the telephone's handset. A copper twisted pair Ethernet cable is usually terminated with a RJ-45 connector. Coaxial cables are normally terminated using a F type connector. A fiber optic cable normally uses a ST, SC, LC, or MTRJ connector.
For support or reporting issues, include Question ID: 63fe05023b7322449ddbafda in your ticket. Thank you.

51 1 / 1 point Multiple Choice

A user reports that their laptop's built-in microphone is not picking up sound during video calls. Which of the following should a technician check first?

- The default playback device settings
- ✓ The application’s microphone permissions
- The microphone hardware connection
- The system's audio driver status

Feedback

General Feedback

OBJ 1.1 - Many applications require explicit permission to access a laptop's built-in microphone, and this should be the first item checked. Default playback device settings affect sound output, not input. Audio driver status could cause microphone issues but is less likely to be the immediate cause if the microphone was working previously. Hardware connection issues would not typically apply to a built-in microphone.
For support or reporting issues, include Question ID: 6797d255aad21b81318d3e37 in your ticket. Thank you.

52 0 / 1 point Multiple Answer

You have been asked by Dion Training to build a network-attached storage (NAS) device that will be used exclusively as a file server for the company's internal network. Which of the following are most important to install to BEST meet the company's needs? (Select TWO.)

- [x] Gigabit NIC
- [ ] RAID 0
- [ ] 1 TB HDD (5400 RPM)
- [ ] RAID 5
- [ ] FastEthernet NIC
- [x] 1 TB SSD

Selected Answer - Incorrect

Feedback

General Feedback

OBJ 3.4: According to CompTIA, the five main things you need to include for network-attached storage (NAS) devices are (1) media streaming capabilities, (2) file-sharing capabilities, (3) Gigabit NIC, (4) RAID array, and (5) hard drives. Based on the options provided, you should pick the RAID 5 and Gigabit NIC for this device's configuration. The RAID 5 will provide both speed and redundancy to the NAS device, whereas the RAID 0 would provide only speed without redundancy. The Gigabit NIC will provide 1 Gbps (1000 Mbps) of network speed instead of the 100 Mbps provided by a FastEthernet NIC. For a NAS device, you should use multiple hard disk drives in a RAID, not a single 1 TB HDD or 1 TB SSD. For a NAS, it is preferred to choose hard drives over solid-state drives because of the increased storage capacity and lower cost.
For support or reporting issues, include Question ID: 63fe05383b7322449ddbb26b in your ticket. Thank you.

53 0 / 1 point Multiple Choice

A technician is troubleshooting a virtual machine that is running in sandbox mode for testing software patches. The user reports that the patch installation caused the VM to crash. What should the technician do next to resolve the issue?

✕ Reset the VM to factory settings and reinstall the patch

Revert the VM to a snapshot taken before the patch was applied

Increase the VM's memory allocation and reinstall the patch

Disable the sandbox mode and apply the patch again

Feedback

General Feedback

OBJ 4.1: The technician should revert the VM to a snapshot taken before the patch was applied. Sandboxes often support snapshots, allowing the system to be restored to a known working state. Resetting the VM or disabling sandbox mode is unnecessary, and memory allocation is unlikely to address the issue directly.
For support or reporting issues, include Question ID: 67987028b1664769adf99b30 in your ticket. Thank you.

54 1 / 1 point Multiple Choice

A customer called the service desk and complained that they could not reach the internet on their computer. You ask the customer to open their command prompt, type in ipconfig, and read you the IP address. The customer reads the IP as 169.254.12.45. What is the root cause of the customer's issue based on what you know so far?

Their workstation cannot reach the web server

Their workstation cannot reach the default gateway

✓ Their workstation cannot reach the DHCP server

Their workstation cannot reach the DNS server

Feedback

General Feedback

<p>OBJ 5.5: Since the customer's IP address is 169.254.12.45, it is an APIPA address. Since the workstation has an APIPA address, it means the DHCP server was unreachable. Automatic Private IP Addressing (APIPA) is a feature of Windows-based operating

systems that enables a computer to automatically assign itself an IP address when there is no Dynamic Host Configuration Protocol (DHCP) server available to perform that function. APIPA serves as a DHCP server failover mechanism and makes it easier to configure and support small local area networks (LANs). If no DHCP server is currently available, either because the server is temporarily down or because none exists on the network, the computer selects an IP address from a range of addresses (from 169.254.0.0 - 169.254.255.255) reserved for that purpose.</p>
For support or reporting issues, include Question ID: 63fe058c3b7322449ddbb679 in your ticket. Thank you.

55 1 / 1 point Multiple Choice

An employee's smartphone is configured via the company's MDM solution. The user complains that they are unable to install a personal messaging app from the app store. You review the MDM policies and find that certain apps are restricted due to corporate policy. The employee insists the app is necessary for their personal use. What is the best course of action?

- ○ Remove the smartphone from the MDM system and allow the employee to install the app freely
- ○ Request IT to whitelist the app in the MDM system to allow installation on all corporate devices
- ○ Temporarily disable the MDM policy on the smartphone so the app can be installed
- ✓ ◉ Explain that the MDM policy blocks certain apps and suggest using the app on a personal device instead

Feedback

General Feedback

OBJ 1.3 - The best course of action is to explain that the MDM policy blocks certain apps and suggest the employee use the app on a personal device instead. MDM policies are implemented to enforce corporate security and compliance, and circumventing or disabling the policy would violate corporate guidelines. Removing the smartphone from the MDM system or whitelisting the app would require approval from IT and may not align with company policy.
For support or reporting issues, include Question ID: 6797dd5b459421aa8e30ca21 in your ticket. Thank you.

56 1 / 1 point Multiple Choice

You are troubleshooting a memory issue on a customer's laptop and have determined that the memory module needs to be replaced. You walk into the storage room to select a memory module to use as a replacement. Which of the following choices would be the MOST LIKELY choice for you to select for use on the laptop?

- ○ Single Channel
- ○ DDR4
- ✓ ◉ SODIMM
- ○ ECC

Feedback

General Feedback

OBJ 3.3: A small outline dual inline memory module (SODIMM) can be purchased in various types and sizes to fit any laptop, router, or other small form factor computing device. For the exam, if you are ever asked about installing memory in a small form factor device or a laptop, the answer will usually be SODIMM. Single-channel memory is any memory using one 64-bit bus between the CPU and the RAM. Error checking and correcting, or error-correcting code (ECC), is a type of system memory that has built-in error correction security. ECC is more expensive than normal memory and requires support from the motherboard. ECC is commonly used in production servers and not in standard desktops or laptops. Double data rate synchronous dynamic (DDR) random access memory is a standard for SDRAM where data is transferred twice per clock cycle to achieve a maximum data rate that is 64 times the bus speed in bps. There are various versions of DDR, including DDR2, DDR3, DDR4, and DDR5.
For support or reporting issues, include Question ID: 63fe04a93b7322449ddbab91 in your ticket. Thank you.

57 0 / 1 point Multiple Choice

You are investigating a network connectivity issue that is affecting two workstations. When you check the network ports that these clients are connected to, you observe that the switch's physical interfaces are continually going up and down. Which of the following is the most likely cause of the port flapping being observed?

× Collisions

Duplicate IP address

Interference

Duplicate MAC address

Feedback

General Feedback

OBJ 5.5: A duplicate MAC address occurs when two or more devices are responding to data requests as if they are the only device on the network with that physical address. One indication of this occurring is when a switch continually changes the port assignments for that address as it updates its content-addressable memory (CAM) table to reflect the physical address and switch port bindings. This will cause the switch ports to continually flap by going up and down as the assignments are updated within the CAM table. A collision is the result of two devices on the same Ethernet network attempting to transmit data at the same time. Collisions are a common occurrence in half-duplex networks but should not occur in a full-duplex switched environment. Interference occurs when electrical signals are sent and received in the same frequency band as your network traffic. Duplicate IP addresses occur when two devices are issued the same IP by a DHCP server or are using identical static IP addresses. Duplicate IP addresses will cause routing issues, but they will not cause port flapping on a switch.
For support or reporting issues, include Question ID: 63fe05673b7322449ddbb4b7 in your ticket. Thank you.

58 0 / 1 point Multiple Choice

Which of the following network performance metrics is used to represent the round-trip time it takes for a packet to be sent by a device to a server and then a response received from that destination server?

Jitter

× Throughput

Latency

Bandwidth

Feedback

General Feedback

OBJ 5.5: Latency is the measure of time that it takes for data to reach its destination across a network. Usually, we measure network latency as the round-trip time from a workstation to the distant end and back. Bandwidth is the maximum rate of data transfer across a given network. Now, bandwidth is more of a theoretical concept that measures how much data could be transferred from a source to a destination under ideal conditions. Throughput is an actual measure of how much data is successfully transferred from the source to a destination. Therefore, we often measure throughput, instead of bandwidth, to monitor our network performance. Jitter is a network condition that occurs when a time delay in the sending of data packets over a network connection occurs. Jitter is a big problem for any real-time applications you may be supporting on your networks, like video conferences, voice-over IP, and virtual desktop infrastructure clients.
For support or reporting issues, include Question ID: 63fe05773b7322449ddbb57a in your ticket. Thank you.

59 1 / 1 point Multiple Choice

The projector in the Dion Training conference room is creating a distorted image when in use. The technician measures the top of the screen at 72” wide and the bottom of the screen at only 66” wide. The technician checks the projector’s resolution and the resolution in the operating system’s display settings. They notice that they are both set correctly to HDTV (1920 x 1080) mode. Which of the following settings on the project should the technician adjust to fix this distortion?

- ✓ (selected) Keystone
- Brightness
- Contrast
- Color depth

Feedback

General Feedback

OBJ 5.3: A keystone effect occurs when the top of a projected image is wider or narrower than the bottom of the image. This creates a trapezoid instead of a rectangular image and leads to distortion. To fix the keystone effect in an image, you need to adjust the keystone setting in the projector. The brightness setting will increase or decrease the

number of lumens of the projected image to make it lighter or darker. The contrast is the amount of difference between the whiteness and darkness in a projected image. Color depth defines how many unique colors can be displayed by the projected image at once. Brightness, contrast, and color depth do not affect the size or shape of the image being projected.
For support or reporting issues, include Question ID: 63fe05973b7322449ddbb705 in your ticket. Thank you.

60 1 / 1 point Multiple Choice

Which of the following technologies can provide up to 40 Gbps of bandwidth?

- USB 2.0
- eSATA
- ✓ Thunderbolt
- FireWire 800

Feedback

General Feedback

OBJ 3.2 - Thunderbolt is the brand name of a hardware interface developed by Intel (in collaboration with Apple) that allows the connection of external peripherals to a computer. Thunderbolt 1 and 2 use the same connector as Mini DisplayPort (MDP), whereas Thunderbolt 3 re-uses the USB-C connector from USB. Thunderbolt 1 reached speeds of 10 Gbps, while Thunderbolt 2 reaches speeds of 20 Gbps. The latest versions, Thunderbolt 3 and 4, can reach speeds of up to 40 Gbps. Thunderbolt is a connector type that can be used either as a display interface (like DisplayPort) or as a general peripheral interface (like USB 3). Thunderbolt 3 uses USB-C connectors. Universal serial bus (USB) is the main type of connection interface used on desktops and laptops. A larger Type A connector attaches to a port on the host; Type B and Mini- or Micro-Type B connectors are used for devices. USB 1.1 supports 12 Mbps while USB 2.0 supports 480 Mbps and is backward compatible with 1.1 devices which run at the slower speed. USB devices are considered hot-swappable and each device can draw up to 2.5 W of power over a USB port. USB 3.0 and 3.1 define 5 Gbps (SuperSpeed) and 10 Gbps (SuperSpeed+) rates and can deliver up to 4.5 W of power. External SATA (eSATA) is widely used to connect hard disks and operates at bandwidths of 1.5 Gbps, 3 Gbps, and 6 Gbps. FireWire is a serial SCSI interface that supports up to 800 Mbps of bandwidth
For support or reporting issues, include Question ID: 63fe052e3b7322449ddbb1f2 in your ticket. Thank you.

61 1 / 1 point Multiple Choice

You have just set up a Minecraft server on a spare computer within your network and want your friends to connect to it over the internet. What do you need to configure in your SOHO router to allow your friends to connect to the new Minecraft server you created?

- Update the firmware
- ✓ Configure port forwarding
- Configure your Wi-Fi to use Channel 11
- Enable DHCP

Feedback

General Feedback

OBJ 2.6: Port forwarding occurs when a router takes requests from the Internet for a particular application, such as HTTP/port 80, and sends them to a designated host on the local area network. This question did not mention that Wi-Fi was being used in the 2.4 GHz frequency range; therefore, using channels 1, 6, and 11 is not required to minimize interference. The dynamic host control protocol (DHCP) is a protocol used to allocate IP addresses to a host when it joins a network. DHCP does not need to be configured to use a server, and most servers use static IP addresses instead of dynamic ones. The firmware is a set of software instructions stored semi-permanently (embedded) on a hardware device. Modern types of firmware are stored in flash memory and can be updated more easily than legacy programmable Read-Only Memory (ROM) types.
For support or reporting issues, include Question ID: 63fe04b93b7322449ddbac5a in your ticket. Thank you.

62 0 / 1 point Multiple Choice

Which storage solution provides the most flexibility and scalability for a virtualized environment?

- Network-attached storage
- ✗ Local solid-state drive
- Direct-attached storage

External USB drive

Feedback

General Feedback

OBJ 4.1: Network-attached storage (NAS) provides a centralized and scalable storage solution that allows multiple virtual machines to access data over the network efficiently. Direct-attached storage (DAS) is directly connected to a single server and lacks flexibility. A local solid-state drive (SSD) offers high speed but lacks network-wide scalability. An external USB drive is not suitable for enterprise-level virtualization due to limited scalability, accessibility, and speed.
For support or reporting issues, include Question ID: 679738e8a7a62dd7fe063cb3 in your ticket. Thank you.

63 1 / 1 point Multiple Choice

What is a tool that would allow you to easily identify a network cable that was not labeled in the wiring closet during its installation?

- Punchdown tool
- ✓ Tone generator and probe
- Cable tester
- Multimeter

Feedback

General Feedback

OBJ 2.8: A tone generator and toner probe is an excellent tool that allows you to easily detect which cable goes to which port on a patch panel in a wiring closet. While you could use a cable tester (if both ends of the cable were available to the technician), a tone generator and probe are much more efficient at identifying cables that were not properly labeled. A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. A cable tester is used to ensure a cable is properly created as a patch cable (straight through) or a crossover cable. Cable testers provide detailed information on the physical and electrical properties of the cable. For example, they test and report cable conditions, crosstalk,

attenuation, noise, resistance, and other cable run characteristics. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack. A multimeter can measure the voltage, amperage, and resistance of a circuit. A multimeter can be a hand-held device useful for basic fault finding and field service work or a bench instrument that can measure electricity with a high degree of accuracy.
For support or reporting issues, include Question ID: 63fe04cf3b7322449ddbad6a in your ticket. Thank you.

64 1 / 1 point Multiple Choice

A technician is building a new computer for Dion Training. The technician installs the motherboard, processor, memory, hard drives, and case fans, and then they attempt to connect the cables to the appropriate connections. When the technician attempts to power on the system, the fans do not spin, the hard drive doesn't spin, and the motherboard's LEDs remain unlit. Which of the following did the technician likely forget to connect to the motherboard?

- ✓ Motherboard 24-pin power cable
- SATA 15-pin power cable
- Case fan 4-pin power cable
- CPU 8-pin power cable

Feedback

General Feedback

OBJ 5.1: Based on the scenario, it is most likely that the technician forgot to connect the motherboard's 24-pin power cable. If the motherboard power cable was not connected, then the motherboard's LED would not be lit. The case fans will not spin either since they receive their power through the motherboard as well. The hard drive will not spin since it receives the wake-up command from the motherboard through a SATA data cable.
For support or reporting issues, include Question ID: 63fe05a53b7322449ddbb7aa in your ticket. Thank you.

65 1 / 1 point Multiple Choice

What is one key characteristic of cloud computing that allows resources to be allocated and

deallocated as needed automatically?

- Availability
- Scalability
- Redundancy
- ✓ Elasticity

Feedback

General Feedback

OBJ 4.2: Elasticity in cloud computing refers to the automatic provisioning and de-provisioning of resources based on demand, ensuring efficient use of resources and cost savings. Scalability refers to the ability to handle an increase in workload over time but does not necessarily imply automatic adjustments. Redundancy refers to data backup, and fault tolerance and availability relate to the reliability and uptime of services rather than dynamic resource adjustment.
For support or reporting issues, include Question ID: 679734dca7a62dd7fe063c7f in your ticket. Thank you.

66 1 / 1 point Multiple Choice

Dion Training wants to install a network cable to run a television signal between two buildings. Which of the following cable types should it utilize?

- Fiber
- Plenum
- ✓ Coaxial
- Shielded

Feedback

General Feedback

OBJ 3.2 - A coaxial cable is a cable used to transmit video, communications, and audio. A coaxial or coax cable is most commonly used as a cable to connect a TV to a cable

TV service. Coaxial cables are a specialized type of copper cabling that uses a copper core to carry the electrical signal while being enclosed by plastic insulation and shielding to protect the data transmission from the effects of electromagnetic interference (EMI). Plenum-rated cable has a special insulation that has low smoke and low flame characteristics. Plenum cable is mandated to be installed in any air handling space. Shielded cables contain a braided foil shield around the inner cabling to protect the data from the effects of electromagnetic interference (EMI). An optical fiber consists of an ultra-fine core of glass to carry the light signals surrounded by glass or plastic cladding, which guides the light pulses along the core, and a protective coating. The fiber optic cable is contained in a protective jacket and terminated by a connector. Fiber optic cables use light signals to carry data across a cable at extremely high bandwidths.
For support or reporting issues, include Question ID: 63fe05283b7322449ddbb1a7 in your ticket. Thank you.

67 1 / 1 point Multiple Choice

Which of the following is used to capture the logs from different devices across the network to correlate different events across multiple servers and clients?

- [x] Syslog server
- [ ] Proxy server
- [ ] DHCP server
- [ ] DNS server

Feedback

General Feedback

OBJ 2.3: A Syslog server is used to capture logs from different devices. It allows for the correlation of logs to simplify log review and an analyst's ability to respond to alerts. For example, Syslog messages can be generated by Cisco routers and switches, servers and workstations, and collected in a central database for viewing and analysis. A proxy server is a server application that acts as an intermediary between a client requesting a resource and the server providing that resource. The Domain Name System (DNS) uses port 53 and is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network. The Dynamic Host Configuration Protocol (DHCP) uses port 67 and is a network management protocol used on Internet Protocol (IP) networks for automatically assigning IP addresses and other communication parameters to devices connected to the network using a client-server architecture.
For support or reporting issues, include Question ID: 63fe04d43b7322449ddbadad in

your ticket. Thank you.

68 0 / 1 point Multiple Choice

Which of the following technologies is commonly used to inventory the organization's equipment by attaching a tag to the device containing a chip programmed with the asset's data, allowing one-way communication to occur from the tag to the reader?

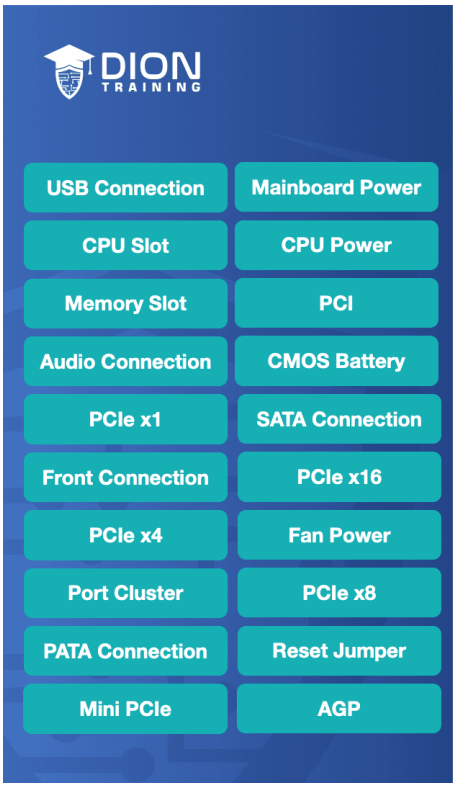
- Zigbee
- RFID
- Z-Wave
- ✕ NFC

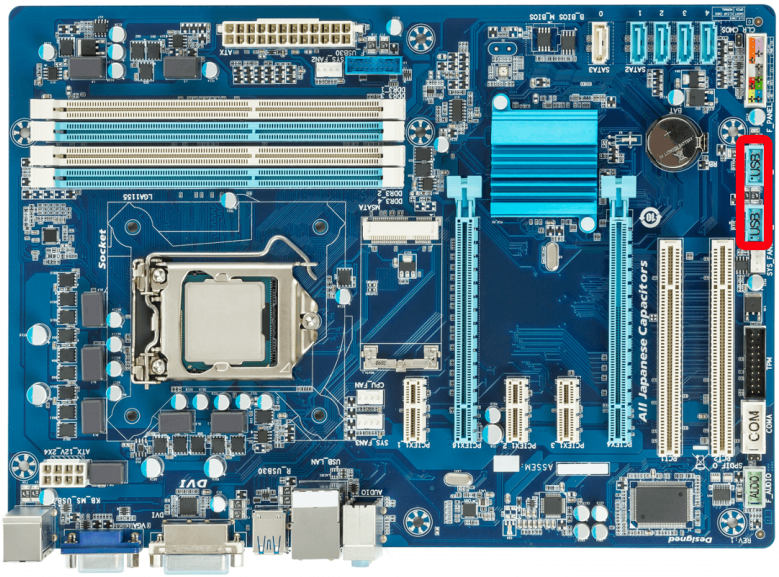
Feedback

General Feedback

OBJ 2.2: An RFID tag contains an RFID chip programmed with the asset data. This is read wirelessly by an RFID (radio frequency identification) reader. RFID is a one-way communication mechanism, whereas NFC is two-way communication. Near-field communication (NFC) is a set of communication protocols that enable two electronic devices, one of which is usually a portable device such as a smartphone, to establish communication by bringing them within 4 cm of each other. This is commonly used for contactless payment systems, transferring contacts, or transferring a file from one device to another. Zigbee is a low-power wireless communications open-source protocol used primarily for home automation. Zigbee uses radio frequencies in the 2.4 GHz band and a mesh topology. Z-wave is a low-power wireless communications protocol used primarily for home automation. Z-wave uses radio frequencies in the 800 MHz and 900 MHz frequency ranges.
For support or reporting issues, include Question ID: 63fe04d43b7322449ddbada3 in your ticket. Thank you.

69 1 / 1 point Multiple Choice





Using the image of the motherboard above, which of the following is circled?

(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- ◯ PS/2
- ✓ ◉ USB
- ◯ SPDIF
- ◯ FireWire

Feedback

General Feedback

OBJ 3.2 - This port is the USB connection. Universal serial bus (USB) is the main type of connection interface used on desktops and laptops. A larger Type A connector attaches to a port on the host; Type B and Mini- or Micro-Type B connectors are used for devices. USB 1.1 supports 12 Mbps while USB 2.0 supports 480 Mbps and is backward compatible with 1.1 devices which run at a slower speed. USB devices are considered hot-swappable and each device can draw up to 2.5 W of power over a USB port. USB 3.0 and 3.1 define 5 Gbps (SuperSpeed) and 10 Gbps (SuperSpeed+) rates and can deliver up to 4.5 W of power. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.

For support or reporting issues, include Question ID: 63fe05223b7322449ddbb15c in your ticket. Thank you.

70 1 / 1 point Multiple Choice

You are troubleshooting a laptop keyboard that has several non-responsive keys. What is the most appropriate first step?

- ◯ Install a virtual keyboard as a temporary fix
- ◯ Replace the entire keyboard assembly
- ✓ ◉ Use compressed air to clean debris from the keyboard
- ◯ Update the keyboard driver in the operating system

Feedback

General Feedback

OBJ 1.1 - The most common cause of non-responsive keys is dirt or debris lodged beneath them. Using compressed air is a quick, non-invasive method to clear this debris and restore functionality. Replacing the keyboard or updating drivers are more time-consuming steps, and installing a virtual keyboard is only a workaround, not a solution. For support or reporting issues, include Question ID: 6797d2f9aad21b81318d3e3c in your ticket. Thank you.

71 1 / 1 point Multiple Choice

A user reports that their mobile device is unable to connect to a corporate Wi-Fi network after switching from their home network. What should you check first to resolve the issue?

- ◯ Reinstall the device's operating system
- ◯ Ensure Bluetooth is disabled on the device
- ◯ Reset the mobile device to factory settings
- ✓ ◉ Verify that the correct SSID and password are entered

Feedback

General Feedback

OBJ 1.3: Since they just switched from their home network, their Wi-Fi is most likely still enabled, so the next step in troubleshooting Wi-Fi connectivity issues is to confirm that the correct network name (SSID) and password are entered. Incorrect credentials are a common cause of connection failures. Checking for Bluetooth connectivity would not help correct the inability to connect to a Wi-Fi network. Other steps, such as factory resets, should be considered only if simpler solutions fail due to the amount of time required to perform the process, when a simpler method may have been adequate.
For support or reporting issues, include Question ID: 67973b2822cbe20a3b2640a3 in your ticket. Thank you.

72 1 / 1 point Multiple Choice

The printer in the Dion Training offices routinely has issues anytime there is a power outage. The printer is a multi-function, network-enabled printer that connects directly to the network switch using a CAT 5e cable. Which of the following should you do to resolve this continuing issue?

- ✓ (selected) Configure a static IP for the printer
- Update the print drivers on each employee's machine
- Disable the SSID broadcast on the wireless access point
- Rollback the print drivers on the print server

Feedback

General Feedback

OBJ 5.5: Since the problem always occurs after a power outage, the printer is likely turning on faster than the DHCP server. This could lead to the printer not getting a new IP address properly or the clients not locating the printer on the network. Therefore, you should configure a static IP for the printer (and any servers) on the network. Servers and shared printers should use a static IP instead of a dynamic assignment to prevent issues.

For support or reporting issues, include Question ID: 63fe05743b7322449ddbb552 in your ticket. Thank you.

73 0 / 1 point Multiple Choice

What is the simplest and most cost-efficient way to allow other network users to access a printer if the printer does not have an RJ-45 or WiFi adapter available?

- Order a printer port converter to share a non-networked printer
- ✕ Establish a printer share on the network switch
- Establish a print share on the local host
- Order another printer with network capabilities and share it on the network

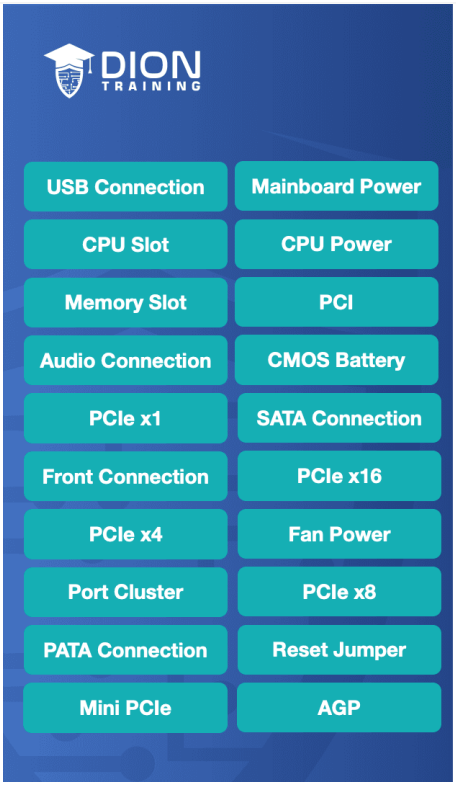
Feedback

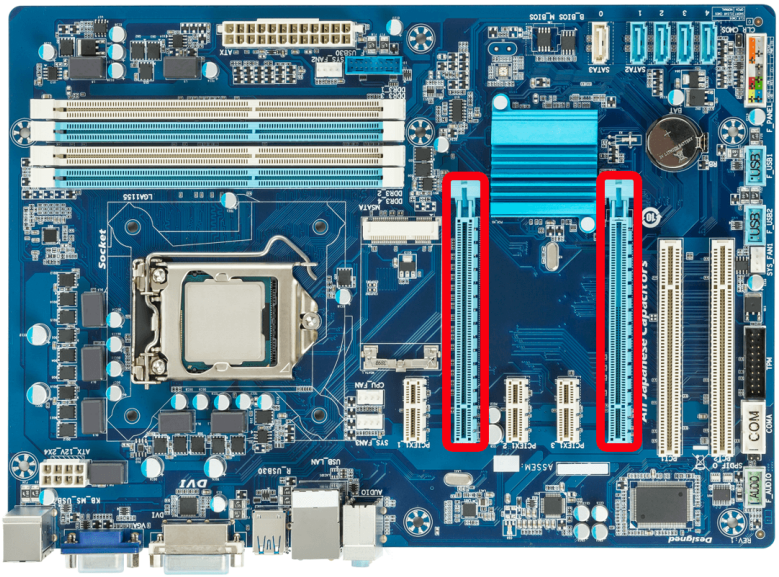
General Feedback

OBJ 5.6: The simplest and most cost-efficient way to share a printer is to share the printer via the local host and allow sharing so everyone can access the printer. While this costs nothing to establish and requires no additional hardware, it does require that the local host remain on 24 hours a day so that others can print to it at any time. If you are willing to spend a little money, though, you can purchase a wireless print server or a new network-capable printer instead.
For support or reporting issues, include Question ID: 63fe05603b7322449ddbb462 in your ticket. Thank you.

74 1 / 1 point Multiple Choice

Using the image of the motherboard, which of the following correctly indicates the circled area?





(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the motherboard.)

- ◯ PCIe x4
- ✓ ◉ PCIe x16
- ◯ AGP
- ◯ PCI

Feedback

General Feedback

OBJ 3.5: The area circled indicates the PCIe x16 (PCI Express x16) slots. PCIe (peripheral component interconnect express) is an interface standard for connecting high-speed components. Every desktop PC motherboard has some PCIe slots you can use to add GPUs (video cards or graphics cards), RAID cards, network adapters, Wi-Fi cards, or SSD (solid-state drive) add-on cards. The types of PCIe slots available in your PC will depend on your motherboard and are designated as PCIe x1, x4, x8, and x16. Graphics cards almost exclusively rely on PCIe x16 expansion slots for their connectivity. For the exam, it is important that you can identify the different parts of the motherboard by sight.
For support or reporting issues, include Question ID: 63fe050d3b7322449ddbb061 in your ticket. Thank you.

75 1 / 1 point Multiple Choice

You are looking for a device that will proactively stop outside attackers from reaching your internal network. Which of the following devices would be the BEST for you to select?

- IDS
- Proxy server
- ✓ IPS
- Syslog server

Feedback

General Feedback

OBJ 2.5: An intrusion prevention system (IPS) is a form of network security that detects and prevents identified threats. Intrusion prevention systems continuously monitor your network, looking for possible malicious incidents and capturing information about them. An IPS can block malicious network traffic, unlike an IDS, which can only log them. A proxy server is a server application that acts as an intermediary between a client requesting a resource and the server providing that resource. System Logging Protocol (Syslog) uses port 514 and is a way network devices can use a standard message format to communicate with a logging server. It was designed specifically to make it easy to monitor network devices. Devices can use a Syslog agent to send out notification messages under a wide range of specific conditions.
For support or reporting issues, include Question ID: 63fe04b93b7322449ddbac55 in your ticket. Thank you.

76 1 / 1 point Multiple Choice

Which of the following modes of operation will allow a motherboard to create two independent 64-bit pathways to achieve 128-bits of data transfer per transaction from the memory to the CPU?

- Single-channel
- Quad-channel

- [x] Dual-channel
- [ ] Triple-channel

Feedback

General Feedback

OBJ 3.3: Dual-channel memory is a memory controller with two primary pathways through the bus to the CPU to transfer 128-bits of data in each transaction. Single-channel memory is any memory using one 64-bit bus between the CPU and the RAM. Triple-channel memory is any memory using three 64-bit buses between the CPU and the RAM. Quad-channel memory is any memory using four 64-bit buses between the CPU and the RAM.
For support or reporting issues, include Question ID: 63fe05123b7322449ddbb098 in your ticket. Thank you.

77 1 / 1 point Multiple Choice

Your computer keeps losing track of time and the date in the system. Which of the following is MOST likely causing this issue?

- [ ] The memory is corrupted
- [ ] The motherboard's device drivers need to be updated
- [x] The CMOS battery is dead and needs to be replaced
- [ ] The power supply is faulty

Feedback

General Feedback

OBJ 5.1: Most motherboards (especially older ones) have a small, coin-sized battery installed on them to power the CMOS RAM when the computer is powered down. This is where the BIOS settings and date/time are saved when power is removed from a workstation. If a computer continually loses track of the day/time, this is a classic sign that the CMOS battery is dead and needs to be replaced. Please note that some newer motherboards have replaced the need for a CMOS battery by replacing it with an area of nonvolatile memory used instead to store the BIOS settings and the date/time.
For support or reporting issues, include Question ID: 63fe056e3b7322449ddbb507 in

your ticket. Thank you.

78 0 / 1 point Multiple Choice

Which motherboard form factor measures 6.7" x 6.7" (17 cm x 17cm) in size?

- [x] mATX
- [ ] ITX
- [ ] ATX
- [ ] mITX

Feedback

General Feedback

OBJ 3.5: The mITX (Mini-ITX) form factor is 6.7" x 6.7" in size (17 cm x 17cm). The mITX is commonly used in smaller computer systems with a small form factor (SFF) case. These motherboards are usually used in computers designed for passive cooling and a low power consumption architecture, such as a home theater PC system. The mATX (microATX) motherboard's form factor is 9.6" x 9.6" in size (24 cm x 24 cm). The mATX form factor is commonly used in smaller computer systems with a small form factor (SFF) case. An mATX motherboard is backward-compatible with the larger ATX motherboard since it contains the same mounting point locations as the full-size board. An ATX motherboard's form factor is 12" x 9.6" in size (305mm x 244mm). ITX is a series of form factors that began with the mini-ITX, but there is no specific size called ITX.
For support or reporting issues, include Question ID: 63fe05093b7322449ddbb02a in your ticket. Thank you.

79 1 / 1 point Multiple Choice

A student wants to use their workstation to run several virtual machines. The workstation currently has an i7 quad-core processor, 4 GB of DDR3 memory, a 512 GB SSD, and a gigabit network interface card. Which of the following components should the student upgrade to support multiple virtual machines running simultaneously on the Windows workstation?

- [ ] SSD

- [x] RAM
- [ ] CPU
- [ ] NIC

Feedback

General Feedback

OBJ 5.1: This workstation is sufficient to run multiple virtual machines as long as the RAM is upgraded. Windows 10 requires at least 2 GB of memory, so only having 4 GB of memory for the entire system is too little to run multiple VMs. It is recommended to upgrade to at least 16 GB of RAM to allow this workstation to support multiple VMs. A virtualization machine needs a lot of memory, multiple processor cores (this already has 4 cores), adequate storage (512 GB of SSD is adequate), and a high-speed network card (1 Gbps is sufficient).
For support or reporting issues, include Question ID: 63fe058f3b7322449ddbb697 in your ticket. Thank you.

80 1 / 1 point Multiple Choice

Which of the following USB connectors is reversible and commonly found on most modern mobile devices?

- [ ] USB Type-A
- [ ] MicroUSB
- [x] USB-C
- [ ] MiniUSB

Feedback

General Feedback

OBJ 1.2: USB-C is reversible, allowing it to be plugged in either way and supports high-speed data transfer and power delivery. USB Type-A is not reversible and is mostly used for computers and peripherals. MicroUSB and MiniUSB are older, non-reversible connectors used on older devices.
For support or reporting issues, include Question ID: 67973274a7a62dd7fe063c66 in

your ticket. Thank you.

81 1 / 1 point Multiple Choice

You have been asked to install a new wireless network for your company's new branch office. Your boss wants the network to be high-speed and is willing to buy new hardware and network adapters to ensure all the devices can communicate on the new network. Which of the following wireless standards should you install for the fastest data transfer speeds?

- [x] 802.11ac
- [ ] 802.11g
- [ ] 802.11a
- [ ] 802.11n

Feedback

General Feedback

802.11ac is the fastest of the four standards listed in this question. The 802.11ac (Wireless AC or Wi-Fi 5) standard utilizes a 5 GHz frequency to provide wireless networking at theoretical speeds up to 3.5 Gbps. Wireless AC uses channel bonding to create a single channel of up to 160 MHz to provide additional bandwidth. Wireless AC uses multi-user multiple-input-multiple-output (MU-MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. The 802.11b (Wireless B) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 11 Mbps. The 802.11g (Wireless G) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 54 Mbps. The 802.11n (Wireless N) standard utilizes a 2.4 GHz frequency to provide wireless networking at speeds up to 108 Mbps or a 5.0 GHz frequency to provide wireless networking at speeds up to 600 Mbps. Wireless N supports the use of multiple-input-multiple-output (MIMO) technology to use multiple antennas to transmit and receive data at higher speeds. Wireless N supports channel bonding by combining two 20 MHz channels into a single 40 MHz channel to provide additional bandwidth.
For support or reporting issues, include Question ID: 63fe04e53b7322449ddbae75 in your ticket. Thank you.

82 1 / 1 point Multiple Choice

Jason's cellphone company advertises that they offer unlimited data. The internet at Jason's

studio was not working today, so he turned on his mobile hotspot to upload the videos recorded today. Initially, the data transfer was occurring without any issues and Jason observed that his data was uploading at 25 Mbps. After 20-30 minutes, though, the speed of the connection dropped to only 0.125 Mbps. Which of the following is the MOST likely cause for this drop in speed during the upload process?

- ◯ The smartphone is undergoing a DDoS attack
- ◯ The smartphone's 802.11 radio is broken
- ◯ The smartphone's battery has been depleted
- ✓ ◉ The smartphone reached the data cap for the month

Feedback

General Feedback

OBJ 1.3: A data cap or bandwidth cap is a service-provider imposed limitation on the amount of data transferred by a user's account or device over the cellular network. Many smartphone plans that advertise unlimited data still have a transfer limit. Once the limit is reached, these plans will either prevent all data transfer from occurring or slow down the speeds of data transfer. For example, T-Mobile offers unlimited data in the United States, but if a user transfers over 50GB of data then their data is deprioritized and transmitted at slower speeds. Other carriers will use a data throttling method, instead, which slows down all data sent for the remainder of the customer's billing cycle. A distributed denial of service (DDoS) attack is a denial of service attack that uses multiple compromised computers, called zombies, to launch the attack against a target. Based on the scenario, there is no evidence that the smartphone's battery is depleted since the phone is still online and transmitting data. The smartphone is not using Wi-Fi to transmit the data and there is no evidence that the 802.11 (Wi-Fi) radio is broken.
For support or reporting issues, include Question ID: 63fe049c3b7322449ddbaaf1 in your ticket. Thank you.

83 0 / 1 point Multiple Choice

Dion Training has a single switch that services every room within its offices. The switch contains 48 ports, but Jason wants to divide the ports based on functional areas, such as web development, instruction support, and administration. Which technology should Jason utilize to divide the physical switch into three logically divided areas?

- ◯ VPN
- ◯ DSL

- VLAN
- ✕ NAT

Feedback

General Feedback

OBJ 2.4: A VLAN (virtual LAN) allows a single physical switch to be divided into logical networks. VLANs are only supported on managed switches, but they allow for a different logical subnetwork address to be assigned to various ports on the switch. This requires that communications between different VLANs must go through a router, just as if you had multiple switches. A virtual private network (VPN) extends a private network across a public network and enables users to send and receive data across shared or public networks as if their computing devices were directly connected to the private network. A digital subscriber line (DSL) modem is a device used to connect a computer or router to a telephone line, which provides the digital subscriber line service for connection to the Internet. Dynamic NAT is a many-to-one mapping of a private IP address or subnets inside a local area network to a public IP address or subnet outside the local area network. The traffic from different zones and subnets over trusted (inside) IP addresses in the LAN segment is sent over a single public (outside) IP address.
For support or reporting issues, include Question ID: 63fe04c13b7322449ddbacbb in your ticket. Thank you.

84 1 / 1 point Multiple Choice

You are trying to select the best device to install to detect an outside attacker trying to reach into your internal network. The device should log the event, but it should not take any action to stop it. Which of the following devices would be the BEST for you to select?

- Authentication server
- ✓ IDS
- Proxy server
- IPS

Feedback

General Feedback

OBJ 2.5: An intrusion detection system is a device or software application that monitors a network or system for malicious activity or policy violations. Any malicious activity or violation is typically reported to an administrator or collected centrally using a security information and event management system. Unlike an IPS, which can stop malicious activity or policy violations, an IDS can only log these issues and not stop them. An intrusion prevention system (IPS) conducts the same functions as an IDS but can also block or take actions against malicious events. An authentication, authorization, and accounting (AAA) server is a server used to identify (authenticate), approve (authorize), and keep track of (account for) users and their actions. AAA servers can also be classified based on the protocol they use, such as a RADIUS server or TACACS+ server. A proxy server is a server that acts as an intermediary between a client requesting a resource and the server that provides that resource. A proxy server can be used to filter content and websites from reaching a user.
For support or reporting issues, include Question ID: 63fe04e63b7322449ddbae7f in your ticket. Thank you.

85 1 / 1 point Multiple Choice

Heath is a user who has submitted a report that they are unable to pair their Bluetooth headphones with their smartphone. Upon inspection, you notice that the headphones are in pairing mode, but the smartphone does not detect them. Which of the following should you advise next when attempting to resolve the issue?

- [x] Ensure that Bluetooth is enabled and scan for available devices
- [ ] Restart the smartphone and attempt pairing again
- [ ] Ensure that the smartphone's operating system has been updated to the latest version
- [ ] Reset the headphones to factory settings and try pairing again

Feedback

General Feedback

OBJ 1.3: The first step is to ensure that Bluetooth is enabled on the smartphone and scan for available devices, as pairing cannot occur if Bluetooth is turned off or if the device is not actively searching. Restarting the smartphone might help but should be attempted after confirming that Bluetooth is enabled. Resetting the headphones should be considered only if other troubleshooting steps fail. Updating the smartphone OS can address compatibility issues but is not the immediate solution to connectivity problems.
For support or reporting issues, include Question ID: 679736caa7a62dd7fe063c8e in your ticket. Thank you.

86 1 / 1 point Multiple Choice

What type of cloud service would provide you with a complete development and deployment environment in the cloud for you to create customized cloud-based apps?

- ◯ DaaS
- ◯ SaaS
- ✓ ◉ PaaS
- ◯ IaaS

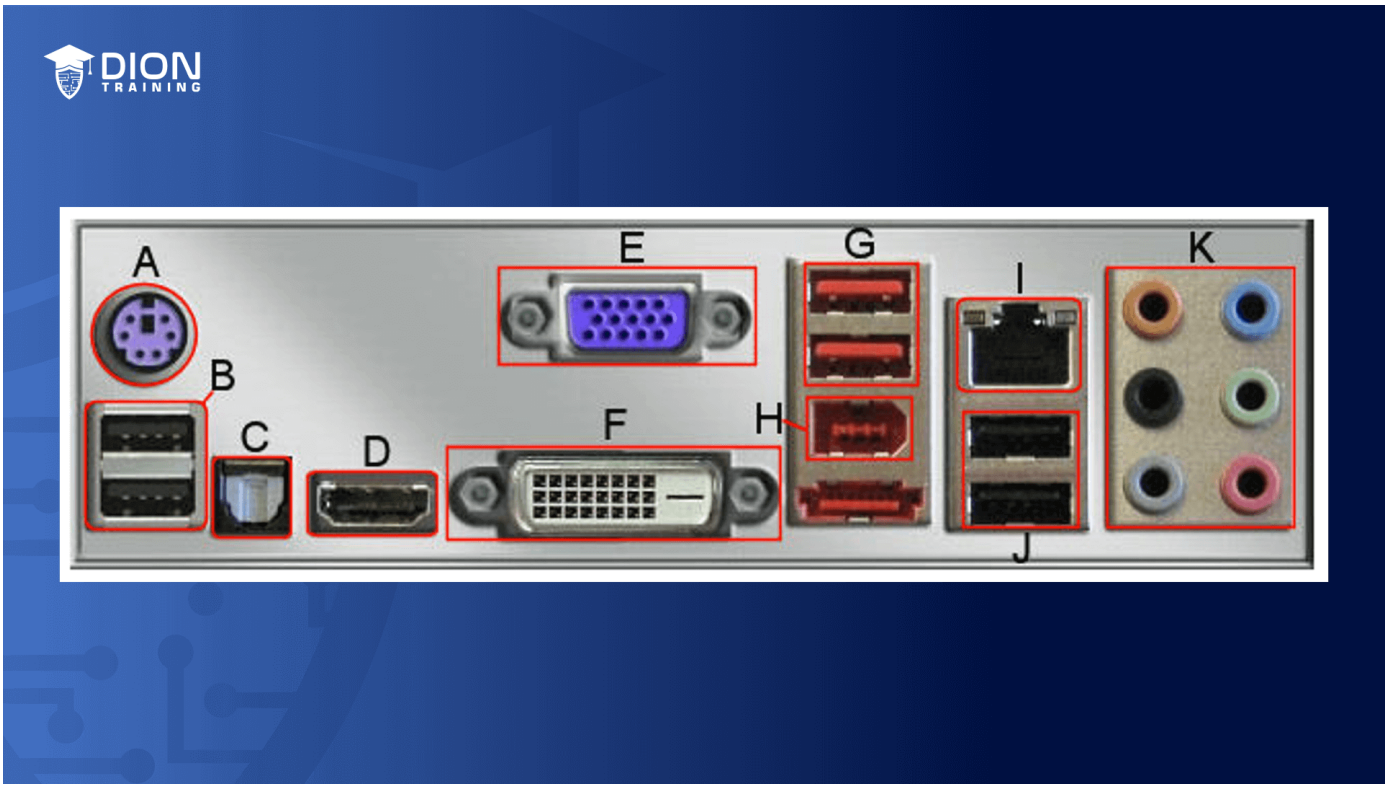
Feedback

General Feedback

OBJ 4.2: Platform as a service (PaaS) is a complete development and deployment environment in the cloud, with resources that enable you to deliver everything from simple cloud-based apps to sophisticated, cloud-enabled enterprise applications. You purchase the resources you need from a cloud service provider on a pay-as-you-go basis and access them over a secure Internet connection. PaaS includes infrastructure (servers, storage, and networking) and middleware, development tools, business intelligence (BI) services, database management systems, and more. PaaS allows you to avoid the expense and complexity of buying and managing software licenses, the underlying application infrastructure and middleware, container orchestrators, or the development tools and other resources. You manage the applications and services you develop, and the cloud service provider typically manages everything else. Infrastructure as a Service (IaaS) is a cloud computing service that enables a consumer to outsource computing equipment purchases and running their own data center. Software as a Service (SaaS) is a cloud computing service that enables a service provider to make applications available over the Internet to end-users. This can be a calendar, scheduling, invoicing, word processor, database, or other programs. For example, Google Docs and Office 365 are both word processing SaaS solutions. Desktop as a Service (DaaS) provides a full virtualized desktop environment from within a cloud-based service. This is also known as VDI (Virtualized Desktop Infrastructure) and is coming in large enterprise businesses focused on increasing their security and minimizing their operational expenses.
For support or reporting issues, include Question ID: 63fe053d3b7322449ddbb2ab in your ticket. Thank you.

87 1 / 1 point Multiple Choice

Using the image, which of the following correctly indicates the port labeled as D?



(This is a simulated PBQ. If you had this question on the real exam, you would be asked to drag-and-drop a label with the component's name onto the proper location on the port cluster.)

- VGA
- DVI-I
- DVI-D
- HDMI

Feedback

General Feedback

OBJ 3.2 - This port is an HDMI connection. A high-definition multimedia interface (HDMI) connection is used to provide high-specification digital content to audio-video equipment from a computer. For the exam, it is important that you can identify the different parts of a motherboard's rear port cluster by sight.
For support or reporting issues, include Question ID: 63fe04fc3b7322449ddbaf92 in your ticket. Thank you.

88 1 / 1 point Multiple Choice

Which of the following synchronization methods does NOT work with Android devices?

- Google Workspace
- Office 365
- Flickr
- ✓ iTunes

Feedback

General Feedback

OBJ 1.3: Android can not use iTunes to perform synchronization functions. The iTunes application only works with Apple devices, such as iPhones and iPads. Android devices can use Google Workspace with Gmail to sync emails and contacts. Android devices can also use Office 365 to sync emails, contacts, and other files if you are a Microsoft subscriber. Android devices can use Flickr to sync photos.
For support or reporting issues, include Question ID: 63fe04963b7322449ddbaaab in your ticket. Thank you.

89 1 / 1 point Multiple Choice

Dion Training's service desk has received calls from numerous users stating they are receiving an error message concerning duplicate IP addresses being detected on their systems. These users also cannot access the network's file server or browse any websites. Which of the following servers should the technician check first in troubleshooting this issue?

- Proxy server
- ✓ DHCP server
- RADIUS server
- DNS server

Feedback

General Feedback

OBJ 2.4: If duplicate IP addresses are being detected on workstations across the network, it is either an issue with the DHCP server or someone has been statically assigning IP addresses to workstations improperly. A dynamic host configuration protocol (DHCP) server is a server configured with a range of addresses to lease. Hosts can be allocated an IP address dynamically or be assigned a reserved IP address based on the host's MAC address. The server can also provide other configuration information, such as the location of DNS servers. A domain name system (DNS) server is a server that hosts the database of domain names and the IP addresses mapped to those names. Each DNS server is authoritative for certain domain names that their organization owns. A proxy server is a server that acts as an intermediary between a client requesting a resource and the server that provides that resource. A proxy server can be used to filter content and websites from reaching a user. An authentication, authorization, and accounting (AAA) server is a server used to identify (authenticate), approve (authorize), and keep track of (account for) users and their actions. AAA servers can also be classified based on the protocol they use, such as a RADIUS server or TACACS+ server.
For support or reporting issues, include Question ID: 63fe04e33b7322449ddbae61 in your ticket. Thank you.

90 1 / 1 point Multiple Choice

You are troubleshooting a network connection issue between the wall jack in a user's office and the switch in the communications closet. When you plug a network cable into the wall jack, there is no connection. You check the switch in the communications closet to determine if the wall jack is properly connected to the switch. Unfortunately, none of the cables in the communication closet are labeled. Which of the following tools should you use to determine which cable in the communications closet is connected to the wall jack in the user's office since none of the cables are properly labeled?

- ✓ (selected) Tone generator and probe
- Multimeter
- Punchdown tool
- Loopback plug

Feedback

General Feedback

OBJ 5.5: A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by detecting the signal being sent by the tone generator. The probe needs to be near

or touching the cable with the tone generator attached to identify it positively. While a multimeter could be used (in resistance mode) to determine if two ends of a cable are attached to the same cable, the distance between the user's office and the communication closet would prevent a multimeter from being used in this case. A tone generator is connected to a wall jack and sends a repeating signal over the cable. The probe can then be used to detect which cable is attached to the wall jack by detecting the signal being sent by the tone generator. The probe needs to be near or touching the cable with the tone generator attached to identify it positively. Punchdown tools are used to connect an Ethernet cable to the back of a patch panel, a punchdown block, or the back of a network wall jack.</p>
For support or reporting issues, include Question ID: 63fe04bb3b7322449ddbac6e in your ticket. Thank you.